SKYTECH ACADEMY

ROPE ACCESS TRAINING CENTER

OILFIELD | INDUSTRY | SAFETY | EXPERTISE



HOUSTON, TEXAS

INDUSTRIAL ROPE ACCESS TRAINING CENTER

TRAINING CURRICULUM

ROPE ACCESS TRAINING

SPRAT certification program — Levels 1, 2 & 3 — for work at height in petrochemical, oil & gas, construction, and wind energy.

THE STANDARD IS COMPETENCE.
THE GOAL IS ZERO.

5-DAY CURRICULUM

CLASSROOM DISCUSSION (1–2 HRS)
FOLLOWED BY HANDS-ON PRACTICE

5 DAYS

skytechacademy.org | 985.661.7100 | info@skytechacademy.org

How the Five-Day Curriculum Works

DAY 01

Foundations

— Orientation
— JSA
— equipment
— redundancy
— knots
— basic movement

DAY 02

Core Techniques

— Regulations
— fall protection
— SPRAT certification & grading
— knot passing
— deviations
— re-anchors
— transfers

DAY 03

Rigging & Rescue

— Anchors
— load physics
— hauling / lowering
— rescue progressions

DAY 04

Advanced & Consolidation

— L2 / L3 rigging
— complex rescue
— scenarios
— mock evaluation

DAY 05

Assessment

— Theory checks
— practical assessment
— debrief
— next-step development

Every Day Follows the Same Learning Rhythm

07:30–08:00	**Start-Up**	Check-in · equipment readiness · JSA / hazard review
08:00–10:00	**Classroom**	1–2 hr guided discussion: what / how · why · when · failures · evaluator · Q&A
10:15–12:00	**Demonstration**	Instructor models the day’s techniques and decision-making
12:45–15:30	**Practice**	Progressive coached stations; repetition until safe and controlled
15:45–16:45	**Consolidation**	Rescue / scenario work, remediation and evaluation-style practice
16:45–17:00	**Debrief**	Questions · equipment accountability · next-day preparation

Safety Through Competence

> **"We prepare technicians not just to pass their certification, but to perform confidently and safely in real industrial environments across Houston's petrochemical, oil & gas, construction, and wind energy sectors."**

Safety First

The training environment is treated as a real worksite — daily JSA, mandatory PPE, zero shortcuts.

Competency-Based

We certify demonstrated skill against objective criteria — not hours spent in a seat.

Field-Realistic

Simulations mirror refinery, tower and offshore conditions students will actually face.

Work-Ready

Graduates leave able to perform the work — and to be productive from day one.

Inspection, Maintenance & Market Opportunity

Petrochemical

Tanks, stacks, columns & vessels — continuous turnaround cycles.

Infrastructure

Bridges, dams & transport structures requiring periodic inspection.

Telecom & Towers

Communication masts & high structures needing at-height service.

Commercial

Stadiums, high-rise façades & large venues across the region.

Every sector needs a steady supply of certified technicians — and every certification expires and must be renewed.

3 yr
SPRAT cert cycle

L1–L3
SPRAT levels

Recurring
Refresher demand

Cost Savings & Productivity

62%

lower total project cost

Documented industrial refinery study, rope access vs. scaffolding for the same scope of work.

Faster mobilization also means less plant downtime — often worth more than the labor savings alone.

WHERE THE SAVINGS COME FROM

Reduced material costs

No scaffold tubes, boards, or fittings to buy, transport and store.

Smaller, skilled teams

A handful of certified technicians replace large scaffolding crews.

Faster mobilization

Setup in hours, not days — work starts and finishes sooner.

Lower exposure hours

Fewer total worker-hours at height reduces risk and overhead.

Real-World Applications

Rope access is the **access method** — professional competence comes from combining it with a trade discipline.

Inspection & NDT

Visual, ultrasonic & non-destructive testing of assets at height.

Maintenance

Mechanical & structural maintenance in hard-to-reach locations.

Construction & Install

Installation and assembly tasks on complex structures.

Coating & Blasting

Surface preparation, painting & protective coating support.

CASE STUDY

Refinery Turnaround

Rope access crews completed inspection & maintenance scopes in a fraction of the time and cost of scaffold-based access — returning the unit to production faster.

62%
lower cost

Hours
to mobilize

SKYTECH ACADEMY

ROPE ACCESS TRAINING CENTER

OILFIELD | INDUSTRY | SAFETY | EXPERTISE

DAY 1

FOUNDATIONS, EQUIPMENT & BASIC ROPE SKILLS

Build the safety habits that every later maneuver depends on.



SAFETY FIRST

EQUIPMENT CONFIDENCE

BASIC ROPE SKILLS

PREPARE. CHECK. PERFORM.
THE STANDARD IS COMPETENCE.
THE GOAL IS ZERO.

1–2 Hour Classroom — Foundations, Equipment & Basic Rope Skills

01 Course orientation & JSA

DISCUSS Who can stop the work — and when?

02 Rope-access foundations

DISCUSS What fails if one rope is cut?

03 Harness, connectors & devices

DISCUSS Which harness point is for what?

04 Inspection, kitting & buddy checks

DISCUSS What takes gear out of service?

05 Knots & terminations

DISCUSS Which knot for which function?

06 Basic ascent, descent & changeover

DISCUSS What holds you during a changeover?

INSTRUCTOR NOTE Do not rush the classroom portion. Use trainee questions and site examples to connect each concept to the practical stations that follow.

INSTRUCTOR DISCUSSION

Course Orientation & JSA

DAY 1

WHAT / HOW

Start the course correctly

Confirm paperwork, training-bay rules, stop-work authority, exclusion zones and the day's Job Safety Analysis before any practical activity. Walk the exclusion-zone boundary physically and confirm every trainee has signed and dated the JSA.

WHY

The classroom comes first

Rope-access competence starts with hazard recognition and planning — not with getting on rope. The JSA gives the team a shared mental model before exposure. Most incidents trace to a hazard spotted too late — the JSA moves that recognition to before anyone is on rope.

WHEN / WHEN NOT

Use it every training day

Review hazards whenever tasks, equipment, weather, personnel or rigging change. Never treat the JSA as a one-time signature exercise. Re-brief after every break and any time the crew composition changes.

WHAT GOES WRONG

Paperwork without understanding

Trainees sign without participating, miss changed hazards, or assume the instructor owns all safety decisions. A signature without participation is the most common paperwork failure on real sites.

EVALUATOR'S EYE

What to observe

Can the trainee identify hazards, controls and stop-work triggers in their own words before approaching the system? Listen for specific named hazards and controls — not vague answers like "be careful."

TRAINEE Q&A

"What if I see something unsafe after we signed?"

Stop the activity, make the area safe, update the JSA / control plan and communicate the change before restarting. Stop-work authority belongs to everyone, including the newest trainee.

Rope-Access Foundations

WHAT / HOW

Three operating principles

Teach redundancy, continuous 100% protection during transitions, and controlled movement. Have trainees identify the working system and the independent backup at each step. Redundancy, continuous 100% attachment and control are the three principles every later maneuver is built on.

WHY

Why redundancy matters

The system is designed so one foreseeable failure does not automatically become a fall. Redundancy only works when the backup is truly independent and correctly managed. Independent means a separate rope, separate anchor and separate attachment — no shared failure point.

WHEN / WHEN NOT

Every maneuver, every transition

Apply before releasing any connection, during transfers, obstacle passage, rescues and edge transitions. Never trade redundancy for speed. Apply the one-rope test constantly: if cutting one line would drop you, the system is wrong.

WHAT GOES WRONG

Single-point moments

Disconnecting too early, both systems sharing a common weak point, slack backups and uncontrolled loading. Each of these is automatic-failure territory on assessment day.

EVALUATOR'S EYE

Evaluator focus

The trainee maintains protection without prompting and recognizes when a proposed sequence creates a single-point failure.

TRAINEE Q&A

“Can I disconnect briefly if I’m holding on?”

No. Hand strength or body position is never a substitute for the required protection system. Grip fails first in a fall or a faint — the system must hold without you.

The Rope Access System at a Glance



Two independent ropes, always. A working line carries the technician; a separate backup line with its own device catches a failure.

Harness, Connectors & Devices

WHAT / HOW

Know the specific equipment

Identify harness attachment points, connectors, descenders, ascenders and backup devices. Teach use according to markings and manufacturer instructions. Cover the ventral, sternal, dorsal and lateral points and what each is rated and intended to do.

WHY

Why generic rules are dangerous

Attachment-point ratings, device compatibility and user ranges vary by model and standard. Correct identification prevents misloading and incompatible combinations. Rope diameter, connector shape and device model must all be verified as a compatible set.

WHEN / WHEN NOT

Use only as designed

Match each device and attachment point to its intended function. Do not improvise uses because a component “looks strong enough.” Dorsal is fall-arrest only; ventral and sternal carry the working rope-access load.

WHAT GOES WRONG

Misorientation & incompatibility

Cross-loaded connectors, unlocked gates, reversed devices, incorrect rope diameter and wrong attachment points can defeat the system. Cross-loading a carabiner across its minor axis can cut its strength by more than half.

EVALUATOR'S EYE

Evaluator focus

Correct selection, orientation, locking, rope path, function check and self / buddy verification. Gates locked, devices correctly oriented, and rope threaded the correct way through each one.

TRAINEE Q&A

“Are all harness D-rings the same strength / use?”

No. Use the harness markings and manufacturer instructions; never assume identical function or rating. Lateral D-rings are used as a pair for positioning, never singly.

Personal Rope Access Kit — What Each Item Does



YOUR PERSONAL KIT — SIX CORE ITEMS

1. **Full-body harness**
   Ventral, sternal, dorsal and lateral attachment points

2. **Hand ascender**
   Grips the working line going up — never alone as backup

3. **Helmet with chinstrap**
   Chinstrap fastened in the fall zone

4. **Energy-absorbing lanyard**
   Connects the backup device; limits arrest force

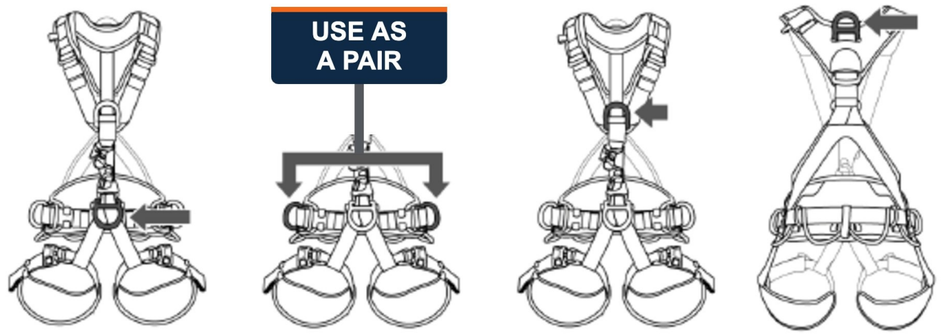
5. **Locking connectors**
   Screw-gate or auto-lock; gate closed and locked

6. **Descender**
   Auto-locking, with panic function where fitted

Example equipment shown. Use the makes and models listed in SkyTech's kit register.

Harness Attachment Points — Four Points, Four Jobs

POINT	LOCATION	PRIMARY USE	MBS	KEY CHECK
Ventral	Front waist	Rope access / positioning — where you live	**5,000 lb**	Gate closed and locked
Sternal	Chest	Backup device · chest ascender · limited fall arrest	**5,000 lb**	Arrest use only if labeled
Dorsal	Between shoulder blades	FALL ARREST ONLY	**5,000 lb**	Buddy checks it for you
Lateral	Side D-rings	Work positioning — always in pairs	**5,000 lb**	Clip both rings together



HARDWARE STRENGTH — MINIMUM BREAKING STRENGTH

D-rings 5,000 lb · Buckles & adjusters 4,000 lb

Dorsal fall arrest limits the force on your body to 1,800 lb. Any ring or buckle that has arrested a fall is removed from service.

STUDENT PRACTICE

Point to and name every attachment point on your own harness; state its MBS and its job before you kit up.

Inspection, Kitting & Buddy Checks

WHAT / HOW

Inspect before use

Check soft goods, hardware, labels, compatibility and function. Fit the harness, install devices, function-test and complete a second-person buddy check. Inspect soft goods for cuts, abrasion, glazing and chemical damage; hardware for cracks, wear and gate action.

WHY

Why two sets of eyes

Routine familiarity creates blind spots. A structured buddy check catches mis-threading, unlocked connectors and incorrect device orientation before exposure. Familiarity breeds blind spots — the buddy check catches what your own eyes skip over.

WHEN / WHEN NOT

Before every practical session

Repeat after equipment changes, breaks, major reconfiguration or whenever there is any doubt. Also re-check after a dropped item or any suspected shock load.

WHAT GOES WRONG

Rushing the check

Skipping labels, overlooking glazing / cuts / corrosion, failing to function-test, or assuming yesterday’s setup is still correct. Yesterday’s “good” is not today’s inspection — verify every session from scratch.

EVALUATOR'S EYE

Evaluator focus

A systematic inspection sequence and willingness to quarantine suspect equipment. Willingness to fail a questionable item is graded as professional judgment, not indecision.

TRAINEE Q&A

“What if damage is questionable?”

Remove it from service and follow the organization / manufacturer inspection process. Never “test it on rope.” Tag it out of service immediately so no one else can pick it up.

The Full Kit — Check It Out, Check It Over

KIT-UP CHECKLIST — BEFORE YOU LEAVE THE STORE

- ☐ **Harness + seat board**
  Fitted, buckles done up
- ☐ **Helmet**
  Chinstrap adjusted
- ☐ **Gloves**
  Good grip, no holes
- ☐ **Chest ascender**
  Mounted, cam clean
- ☐ **Hand ascender + foot loop**
  Length set to your leg
- ☐ **Descender**
  Correct rope diameter
- ☐ **Backup device + absorber**
  Matched as a tested pair
- ☐ **Cowstails / lanyards**
  Stitching and terminations
- ☐ **Connectors**
  Gates close and lock
- ☐ **Slings, pulleys, rope bag**
  Counted and tagged

Buddy check every item before you load a rope.



Inspect Before Use — Ascenders & Descenders



HAND ASCENDER

1. Cam teeth sharp, clean, not worn
2. Safety catch opens fully, snaps back
3. Frame free of cracks, bends, sharp edges
4. Attachment holes not worn or elongated



CHEST ASCENDER

1. Cam springs back and grips the rope
2. Teeth clean — no mud, ice or paint
3. Top and bottom holes not worn sharp
4. Fitted so the rope runs straight up



AUTO-LOCKING DESCENDER

1. Side plate opens, closes and latches
2. Handle returns; anti-panic engages
3. Friction surfaces not deeply grooved
4. Rope path matches the diagram on the device



SIMPLE DESCENDER

1. Side plate and safety catch close fully
2. Handle springs back when released
3. Bollards not deeply grooved or sharp
4. Releasing the handle stops the descent

Function-test at ground level before every climb — load it, release it, and watch it lock.

Inspect Before Use — Harness, Helmet & Backup



HARNESS

1. Webbing: no cuts, burns, fraying or fading
2. Stitching intact — no pulled threads
3. Buckles lock; D-rings not bent or corroded
4. Label legible and within its lifespan



HELMET

1. Shell: no cracks, dents or deep gouges
2. Chinstrap and buckle hold firm
3. Headband and clips intact, adjust easily
4. Retire after any heavy impact



BACKUP DEVICE

1. Runs freely up and down the rope
2. Locks at once when jerked downward
3. Connector captive, gate locked
4. Fitted the right way up — arrow up



ROPE GRAB

1. Cam moves freely; spring returns
2. Grips when loaded, releases cleanly
3. Axle and side plate secure
4. No cracks, bends or heavy wear

Retire it after a fall arrest, at the end of its marked lifespan, or whenever you are in doubt.

Inspect Before Use — Connectors, Rope & Pulleys



SCREWGATE CONNECTOR

1. Gate opens, closes and latches fully
2. Sleeve screws fully shut, no binding
3. No cracks, sharp edges or deep grooves
4. Loaded along the spine, never the gate



AUTO-LOCKING CONNECTOR

1. Gate locks by itself on release
2. Springs shut without sticking
3. Steel body free of rust and bends
4. Never cross-loaded over an edge



ROPE

1. Sheath: cuts, abrasion, glazing
2. Core: flat spots, lumps, stiffness
3. No chemicals, paint or stains
4. Ends and knots sound; logbook updated



PULLEY

1. Sheave turns freely, no wobble
2. Side plates straight, open and close
3. Sheave groove smooth, not worn
4. Rated loads marked and legible

Any doubt? Tag it out of service and follow the manufacturer’s inspection instructions.

Knots & Terminations

WHAT / HOW

Build, dress, set, inspect

Practice the figure-eight family, figure nine, alpine butterfly, stopper / barrel, clove hitch, double fisherman’s and friction hitches as required by the course. For each, teach the correct application, how to dress it and how to inspect it at a glance.

WHY

Why knot quality matters

A correctly selected knot can still be unsafe if poorly dressed, inadequately tailed, loaded incorrectly or used for the wrong purpose. A poorly dressed knot can lose a large fraction of the rope’s strength.

WHEN / WHEN NOT

Select by function

Use knots only where appropriate for attachment, isolation, termination, joining or rigging. Consider load direction and whether it must be untied after loading. Match the knot to the function — not to habit.

WHAT GOES WRONG

Wrong knot / poor dressing

Crossed strands, short tails, incorrect orientation and unverified knots are recurring errors. A missing stopper knot in a rope end is an instant assessment write-up.

EVALUATOR'S EYE

Evaluator focus

Selection, dressing, setting, adequate tails, correct loading and an independent partner check. Tail length of roughly a fist-width is checked on every knot.

TRAINEE Q&A

“Which knot is strongest?”

The better question: which approved knot correctly performs the required function in this system. There is no single “strongest” knot — only the right knot, correctly tied, for the job.

Seven Course Knots — What Each One Looks Like



Figure 8 SPRAT L1

Default attachment & anchor — checkable at a glance



Figure 9 SPRAT L1

Sustained anchor load — keeps more strength, unties easier



Alpine Butterfly SPRAT L1

Mid-line attachment; loads 3 ways; isolates damage



Stopper SPRAT L1

Closes the rope end — can't descend off a closed rope



Double Fisherman's SPRAT L1

Connects two ropes for retrievable rigging



Prusik COURSE SKILL

Friction hitch — not a SPRAT test item



Munter COURSE SKILL

Belay / lower on a carabiner

DRESSING STANDARD

Correctly formed · dressed flat · set snug · fist-width tail.

COMMON MISTAKES

Crossed strands · short tail · wrong knot for the load · missing stopper knot.

Every knot reduces rope strength — how much depends on the knot, rope and dressing.

INSTRUCTOR CHECKPOINT SPRAT L1 tests four functions: termination, middle, stop and join. Each trainee ties, dresses and sets every knot and passes a partner check.

Photos via Wikimedia Commons: WikipedianYknOK (CC BY-SA 3.0) — figure 8, alpine butterfly, double fisherman's, prusik, munter; Markus Bärlocher (public domain) — figure 9, stopper.

Basic Ascent, Descent & Changeover

WHAT / HOW

Move deliberately

Teach efficient ascent, controlled descent, backup management and ascent-to-descent changeovers while maintaining continuous protection. Efficient ascent drives from the legs in the footloop; the arms only guide, never haul.

WHY

Why sequence matters

Device order controls whether the trainee remains supported and protected throughout the transition. Load and check the next device before releasing the current one — every changeover, every time.

WHEN / WHEN NOT

Master before obstacles

Trainees should be stable with basic movement before deviations, re-belays, transfers or rescue tasks. Fluency here is the gate to Day 2’s obstacle work.

WHAT GOES WRONG

Uncontrolled movement

Poor brake-hand discipline, unmanaged slack, releasing a device too early, or loading the wrong system. Hauling with the arms instead of standing in the footloop causes premature exhaustion.

EVALUATOR'S EYE

Evaluator focus

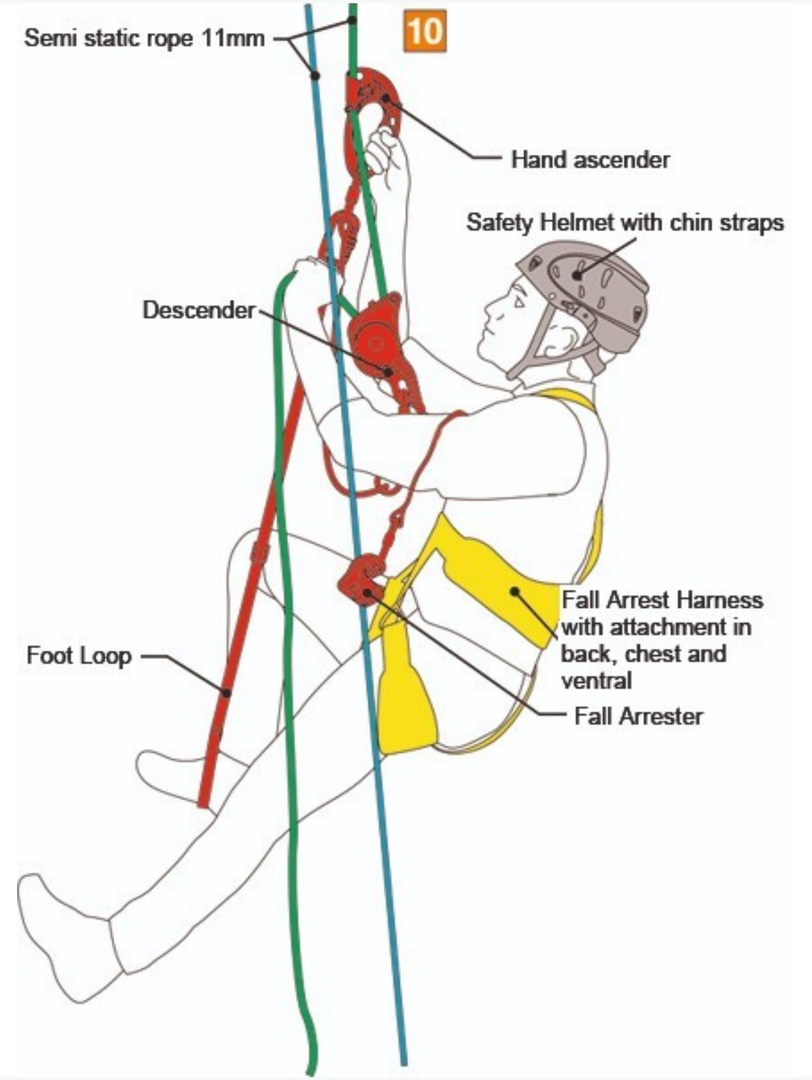
Smooth sequencing, independent backup, controlled body position and the ability to stop safely at any point. Backup kept high and no slack in the system are watched on every meter of rope.

TRAINEE Q&A

“Should I memorize hand movements?”

Learn the sequence and the reason behind it. You must be able to adapt while preserving the safety principles. You should also be able to reverse a changeover mid-sequence if something feels wrong.

Ascent Set-Up — Where Every Device Goes



ASCENT SET-UP — CHECK TOP TO BOTTOM

1. **Two ropes**
   Working line and a separate backup line, each on its own anchor

2. **Hand ascender + foot loop**
   Stand up in the foot loop, slide the hand ascender up

3. **Chest ascender**
   On the working line; takes your weight as you sit back

4. **Descender**
   Stays on the working line, ready to change over

5. **Backup device**
   On the backup line, kept above your attachment point

6. **Helmet and harness**
   Chinstrap on; every connector locked

Before leaving the ground: self-check, buddy check, then a test load close to the ground.

Logbooks, Experience & Recertification

WHAT / HOW

Maintain verifiable experience records

Log every job in your SPRAT logbook — date, employer, task, rope hours — with a verifying signature from your Level 3, program administrator or client.

WHY

Why documentation matters

Upgrade eligibility and professional credibility depend on accurate, auditable experience.

WHEN / WHEN NOT

Record work promptly

Log each job as it happens. Hours worked while your certification is expired do not count toward the next level.

WHAT GOES WRONG

Inflated or vague entries

Unverifiable hours, missing supervisors, unclear scope and incorrect level claims.

EVALUATOR'S EYE

Evaluator focus

Candidates understand that official SPRAT requirements control upgrade / renewal eligibility.

TRAINEE Q&A

“How many hours do I need?”

Level 2: 500 hrs + 6 months as L1. Level 3: 500 hrs + 6 months as L2. Always confirm against the current SPRAT Certification Requirements.

Career Development & Trade Integration

WHAT / HOW

Rope access is the access method

Combine access competence with a technical trade — inspection / NDT, mechanical, coatings, electrical, welding / fitting, wind or another discipline.

WHY

Why trade skill matters

Employers need a task performed at height — not simply a person who can climb ropes.

WHEN / WHEN NOT

Build competence progressively

Choose training that matches the work you intend to do and keep both access and trade qualifications current.

WHAT GOES WRONG

Working outside competence

Rope-access certification does not qualify a person for an unrelated technical trade.

EVALUATOR'S EYE

Evaluator focus

Professional judgment includes recognizing the limits of both rope-access level and trade competence.

TRAINEE Q&A

“What should I learn next?”

Choose the technical discipline that fits your target industry, then build verified experience while keeping rope-access skills current.

CLASSROOM → PRACTICAL

Day 1 Practical Training Plan

- Equipment inspection & buddy-check stations
- Knot tying / dressing / inspection repetitions
- Basic ascent and controlled descent
- Ascent ↔ descent changeovers
- Edge / start-position drills
- End-of-day competency check before Day 2



Photo: Mika343243 · CC BY-SA 4.0 · Wikimedia Commons

SKYTECH ACADEMY

ROPE ACCESS TRAINING CENTER

OILFIELD | INDUSTRY | SAFETY | EXPERTISE

DAY 2

REGULATIONS, FALL PROTECTION & CORE MANEUVERS

Connect standards and fall-protection theory to real movement on rope.



STANDARDS
SPRAT L1 · L2 · L3 REQUIREMENTS

FALL PROTECTION
SYSTEMS & INSPECTION

CORE MANEUVERS
CONTROL.
EFFICIENCY.
CONFIDENCE.

APPLY KNOWLEDGE
PERFORM SAFELY.
WORK SMART.
PROTECT EVERYONE.

REGULATIONS & STANDARDS | FALL PROTECTION SYSTEMS | ROPE HANDLING FUNDAMENTALS | CORE MANEUVERS ON ROPE

KNOW THE RULES.
MASTER THE MOVEMENT.
WORK WITH CONFIDENCE.

1–2 Hour Classroom — Regulations, Fall Protection & Core Maneuvers

01 **Regulatory framework & SPRAT certification**

DISCUSS Law, standard or SPRAT rule — which is it?

02 **Fall-protection systems**

DISCUSS Which system controls this hazard?

03 **Fall factor, clearance & suspension**

DISCUSS How far could you fall here?

04 **Deviations**

DISCUSS What does the deviation anchor carry?

05 **Re-belays / re-anchors**

DISCUSS What protects you mid-re-anchor?

06 **Knot passing & rope-to-rope transfers**

DISCUSS When is the new system ready to load?

INSTRUCTOR NOTE Use trainee questions and site examples to connect each standard and technique to the practical stations that follow.

Regulatory Framework

WHAT / HOW

Know which rule set you are using

Separate legal requirements, consensus standards, certification-scheme rules, employer procedures and manufacturer instructions.

WHY

Why sources must stay distinct

OSHA, ANSI and SPRAT serve different functions. Mixing them can turn guidance into a false legal requirement — or omit an actual one.

WHEN / WHEN NOT

Verify current editions

Use the current source for the work environment and certification scheme. Site / client rules may be more restrictive.

WHAT GOES WRONG

Memorizing numbers without context

Applying one anchor rating, force limit or equipment rule universally is unsafe and often incorrect.

EVALUATOR'S EYE

Evaluator focus

Trainees explain where a requirement comes from and when they must escalate to a supervisor / qualified person.

TRAINEE Q&A

“Does SPRAT replace OSHA?”

No. A certification scheme does not replace applicable law, employer duties or site requirements.

What Is SPRAT?

The **Society of Professional Rope Access Technicians** is a North American organization that publishes rope access safe-practice standards and certifies technicians at three levels.

GOVERNING DOCUMENTS

— Safe Practices for Rope Access Work

— Certification Requirements

— Evaluation Guidelines & Matrix

— Defined Terms

L1 — **Rope Access Technician — Entry**

No experience required · training within 6 months before evaluation · written test + practical.

L2 — **Rope Access Technician — Intermediate**

500 hrs + 6 months as L1 · system analysis · retrievable rigging · cross-haul · fall-arrest rescue.

L3 — **Rope Access Supervisor — Advanced**

500 hrs + 6 months as L2 · tensioned systems · rescue from maneuvers · site supervision.

3 Years
Certification cycle

Age 18+
Minimum entry requirement

1 F or 3 D
Pass / Discrepancy / Fail

SPRAT Evaluation — How Candidates Are Graded

WHAT / HOW

Written test + practical evaluation

Certification requires passing a written test and a practical evaluation run by an independent SPRAT evaluator. The evaluator, not the trainer, has sole authority to issue grades.

WHY

Pass · Discrepancy · Fail

Each exercise is graded P, D or F against the performance principles. One Fail or three Discrepancies fails the evaluation — several Ds can come from one exercise.

WHEN / WHEN NOT

Safe and efficient

Every requirement must be completed safely and efficiently. SPRAT does not specify equipment or technique — the principles are what is graded.

WHAT GOES WRONG

Common discrepancy sources

Backup free fall over 4 ft, main-system free fall over 2 ft, swing fall over 2 ft, no backup in the fall zone, or a single component that could fail and drop the load.

EVALUATOR'S EYE

What the evaluator watches

A backup with every main system, free fall and swing within limits, chinstrap fastened in the fall zone, and no way to become detached.

TRAINEE Q&A

“What happens if I fail?”

The evaluator explains every D or F before your next exercise, and you receive a provisional result. Retrain the flagged skills before re-evaluating.

SPRAT Career Path — Eligibility & Renewal

WHAT / HOW

Level 1 → Level 2 → Level 3

L1: age 18+, government ID, no prior experience. L2: 500 hrs + 6 months as L1. L3: 500 hrs + 6 months as L2. Advancement is consecutive — no skipping levels.

WHY

Each level adds responsibility

L1 masters maneuvers, simple two-rope systems and a pick-off rescue. L2 adds system analysis, retrievable and remote-rescue rigging, cross-hauls and fall-arrest rescue. L3 adds tensioned systems and rescue from maneuvers.

WHEN / WHEN NOT

Renew every 3 years

Certification is valid for three years. Retrain and re-evaluate within the 6 months before expiry and the new certificate runs 3 years from the old expiry date.

WHAT GOES WRONG

Letting certification lapse

Hours worked without a valid certificate do not count toward an upgrade, and expired candidates must demonstrate every skill required at their level.

EVALUATOR'S EYE

Documented, verified hours

Upgrade candidates document 500 hours at their current level, recorded and verified as SPRAT Safe Practices require.

TRAINEE Q&A

“Is training required before evaluation?”

Yes. SPRAT requires training by a trainer within six months before initial certification at each level, and recommends it before any later evaluation.

Fall-Protection Systems

WHAT / HOW

Choose the right control

Discuss elimination / avoidance, restraint, work positioning / rope access, and fall arrest. Identify which system is actually controlling the hazard.

WHY

Why system selection matters

Different systems assume different exposure and loading. Equipment appropriate for positioning may not be a complete fall-arrest solution.

WHEN / WHEN NOT

Before connecting

Identify the hazard, system purpose, anchor strategy, clearance, swing exposure and rescue plan.

WHAT GOES WRONG

Using the wrong system

Insufficient clearance, excessive free fall, poor anchor location, incompatible components and no viable rescue.

EVALUATOR'S EYE

Evaluator focus

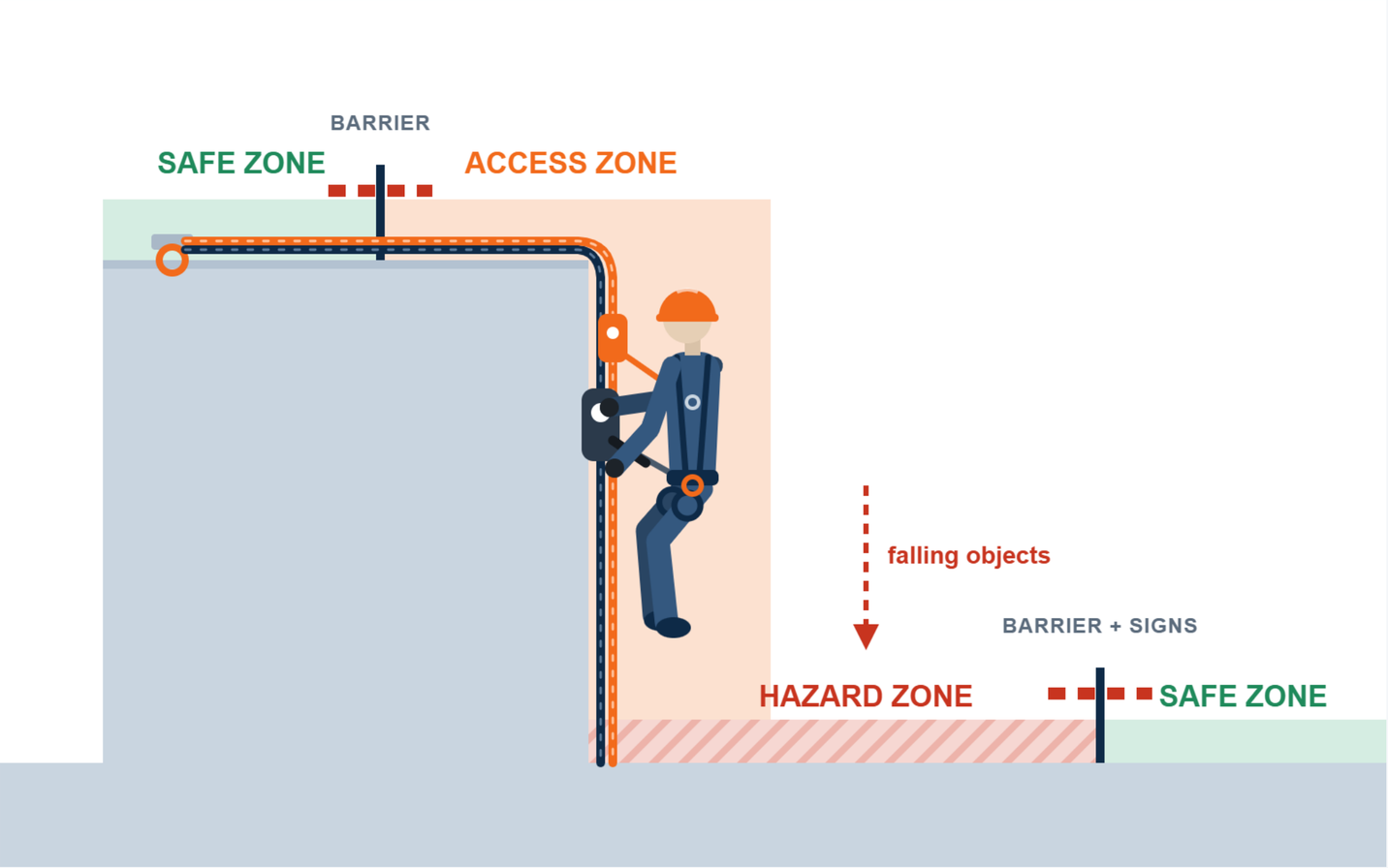
Can the trainee identify the system and explain what prevents or arrests the fall?

TRAINEE Q&A

“If I have a lanyard, am I protected?”

Only if the complete system — connection, anchor, clearance and use — is appropriate for the exposure.

Access, Hazard and Safe Zones



ACCESS ZONE

The controlled area where rope access is taking place, where people are at risk of falling: on rope or near a working edge.

HAZARD ZONE

Areas around the Access Zone where people may be at risk from the work being performed, such as falling objects.

SAFE ZONE

Areas outside the Access and Hazard Zones, safe from the effects of the rope access work. That does not mean hazard-free.

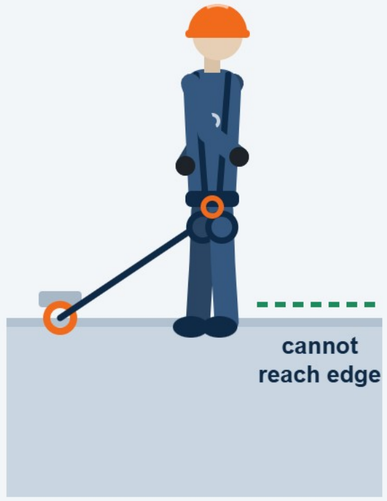
THE RULE

Barricade and sign the Hazard Zone before work starts.

THE REMINDER

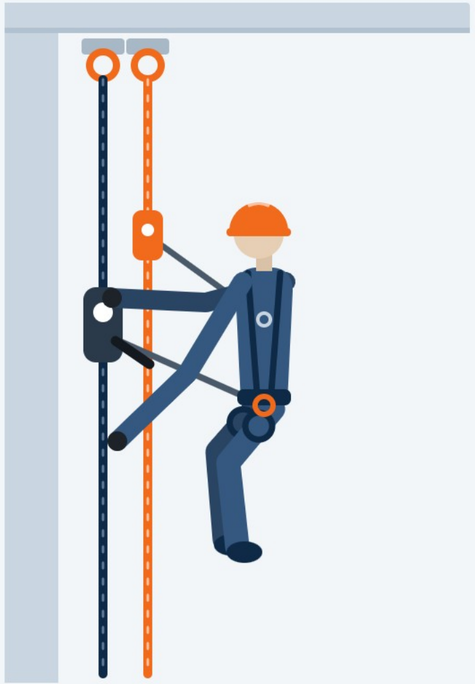
The Safe Zone is safe from the rope access work. It is not free of every hazard.

Four Systems — Prevent First, Arrest Last



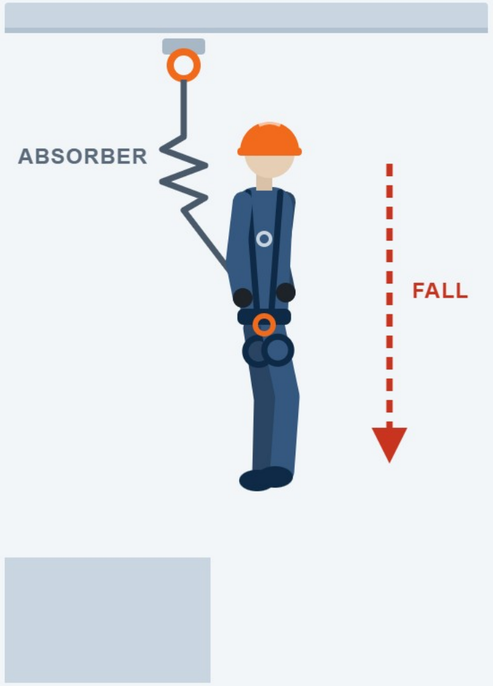
RESTRAINT

The fall is eliminated



WORK POSITIONING

Supported on two ropes



FALL ARREST

Stops a fall already under way



MANAGED PROGRAM

Wraps around all three

THE ORDER

Restraint first, work positioning for rope access, fall arrest only as the last resort.

THE WRAPPER

A managed program covers procedures, training, inspection and rescue for every system.

Fall Factor, Clearance & Suspension

WHAT / HOW

Understand the consequences of a fall

Explain fall potential, energy absorption, clearance components, swing hazards, and the need for prompt rescue after a fall or incapacitation.

WHY

Why calculations matter

A compliant component does not guarantee a safe system if there is insufficient clearance or the geometry creates impact / swing hazards.

WHEN / WHEN NOT

Plan before exposure

Evaluate the complete arrest path and rescue access before work begins.

WHAT GOES WRONG

Generic shortcuts

Do not rely on a universal “20-minute” rule or a simplified clearance guess. Treat motionless suspension as urgent and use current rescue / medical guidance.

EVALUATOR'S EYE

Evaluator focus

Candidate identifies free-fall / clearance concerns and does not proceed when the available space is uncertain.

TRAINEE Q&A

“How much clearance do I need?”

It depends on the complete system. Calculate using the manufacturer / system data plus an appropriate safety margin.

Four Systems & Fall-Arrest Limits

PREVENT · PREFERRED

Restraint

Worker cannot reach the hazard — the fall is eliminated.

SUPPORT · ROPE ACCESS

Work Positioning

Supported for hands-free work; dual-rope redundancy.

LAST RESORT

Fall Arrest

Stops the worker after a fall begins — force reaches the body.

THE WRAPPER

Managed Program

Procedures, training and inspection (ANSI Z359.2) around it all.

CONNECTION	PEAK FORCE	SOURCE	MAX FREE FALL
Sternal (front)	**900 lb · 4 kN**	Harness label	2 ft
Dorsal (back)	**1,800 lb · 8 kN**	OSHA 1926.502	6 ft*

* OSHA also limits deceleration distance to 3.5 ft · always follow the harness label

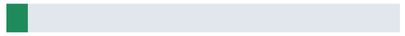
WHY ABSORBERS WORK

Force = mass × deceleration. An absorber tears progressively, stretching the stop over more time — more tear time, less peak force.

Fall Factor & Clearance — the Two Calculations

FF = Fall ÷ Rope

3-ft cowstail — where you keep the backup decides everything.

FF 0		Backup above attachment · 0 ft ÷ 3 ft = 0
FF 1		Backup level with attachment · 3 ft ÷ 3 ft = 1
FF 2		Backup at feet · 6 ft ÷ 3 ft = 2 — over 4-ft limit

Most-watched habit on assessment: SPRAT caps backup free fall at 4 ft (aim ≤ 2 ft) and main-system free fall at 2 ft. Keep the backup high.

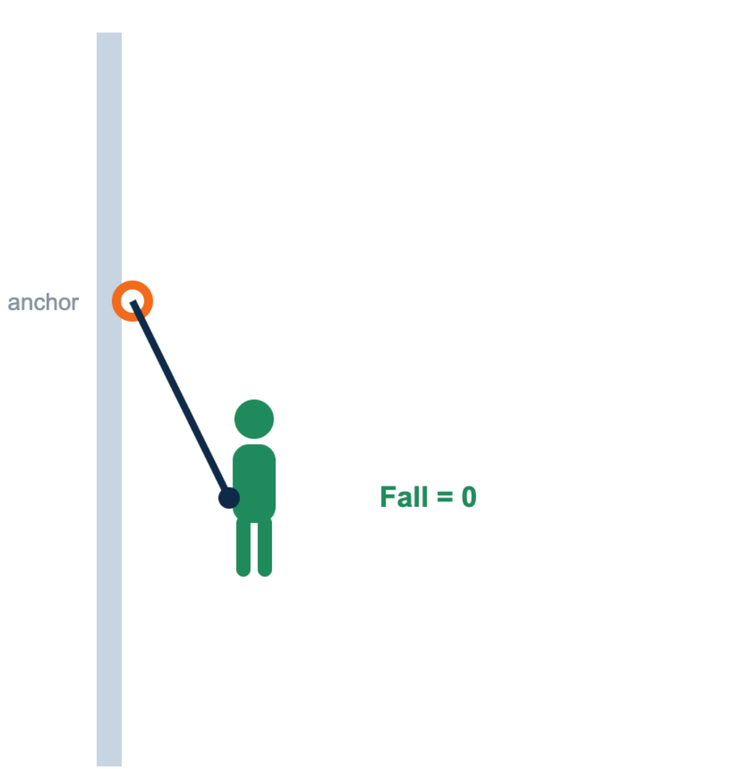
CLEARANCE — WORKED EXAMPLE

6-ft energy-absorbing lanyard, 6-ft worker · example values

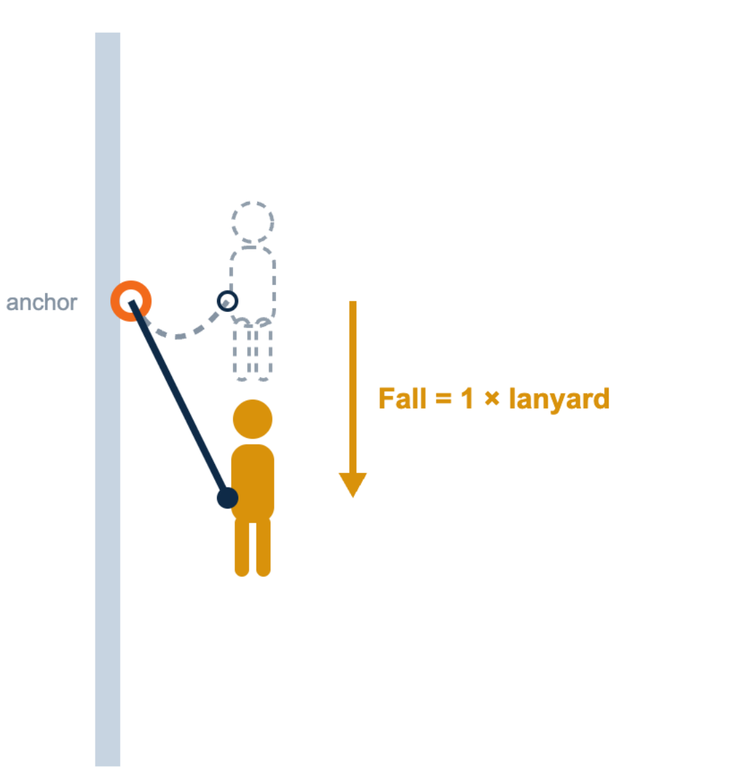
Free fall (lanyard length)	**6.0 ft**
Absorber deployment (OSHA max)	**3.5 ft**
D-ring to feet (worker height)	**5.0 ft**
Harness stretch + D-ring shift	**1.0 ft**
Safety margin	**2.0 ft**
Clearance needed below anchor	**17.5 ft**

The shock: working 15 ft above the ground with this lanyard, you hit the ground before the fall is arrested. Use your own system’s numbers.

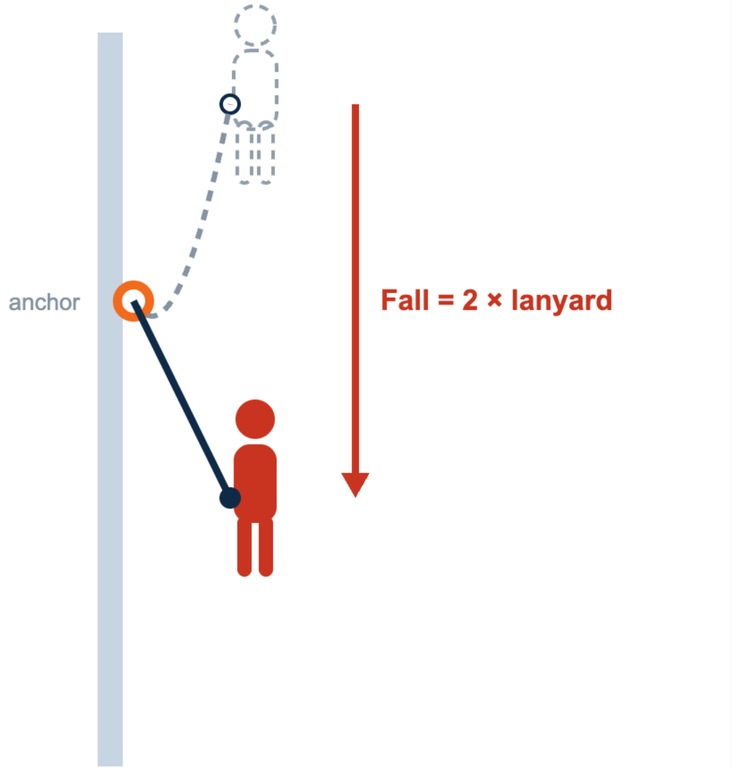
Fall Factor — Where the Backup Sits Decides the Fall



FF 0 · Backup above you



FF 1 · Backup level with harness



FF 2 · Backup below you (at feet)

THE FORMULA

Fall factor = distance fallen ÷ length of lanyard

THE HABIT

Keep the backup above your harness attachment — aim for FF 0 on every move.

Deviations

WHAT / HOW

Redirect the rope path safely

Approach under control, manage both systems, transfer the deviation connection in sequence and control pendulum / swing potential.

WHY

Why deviations are used

They redirect ropes around hazards, improve access geometry, or protect the system without creating a full re-anchor.

WHEN / WHEN NOT

Use when geometry supports it

Consider deviation angle, rescue implications and whether the redirection creates dangerous swing or loading.

WHAT GOES WRONG

Disconnecting in the wrong order

Loss of protection, uncontrolled swing, trapping the device or loading a connector incorrectly.

EVALUATOR'S EYE

Evaluator focus

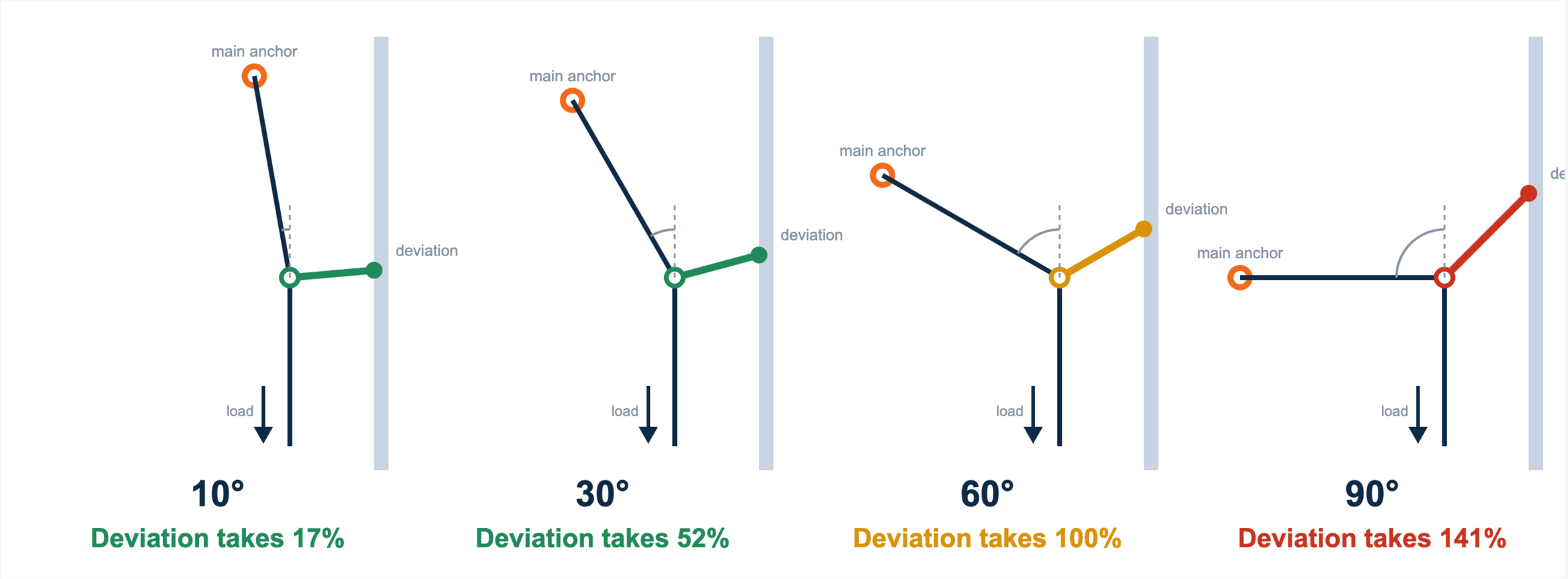
Continuous protection, correct sequence, controlled body position and awareness of swing hazards.

TRAINEE Q&A

“Is a deviation an anchor?”

It redirects the rope path — teach the specific system design; do not assume it functions like a primary independent anchor.

Deviations — The Bigger the Angle, the Bigger the Load



WHY IT MATTERS

Load on the deviation = 2 × sin(½ × angle) × load. A small redirect puts little load on the deviation anchor and sling; at 60° the deviation carries as much as the main anchor, and beyond that it carries more. Choose deviation anchors and slings rated for that load.

Re-belays / Re-anchors

WHAT / HOW

Transfer past a new anchor station

Build / verify the next protected section, transfer devices in a controlled order and never create an unprotected moment.

WHY

Why re-belays are used

They isolate rope sections, manage edges and obstructions, reduce stretch, or create better access geometry.

WHEN / WHEN NOT

Plan the sequence before arriving

Know where each device and connection will go before unloading the existing section.

WHAT GOES WRONG

Crowding and wrong sequence

Loading both systems incorrectly, excessive loop slack, trapped devices, dropped connections or loss of redundancy.

EVALUATOR'S EYE

Evaluator focus

Safe order, controlled loading / unloading, tidy rope management and independent protection throughout.

TRAINEE Q&A

“Which device moves first?”

Follow the approved technique and understand what is supporting / protecting you at every step — do not memorize an isolated trick.

Knot Passing & Rope-to-Rope Transfers

WHAT / HOW

Move around an obstruction or onto another system

Establish the new support / protection before progressively unloading the old system; manage devices so no step creates uncontrolled movement.

WHY

Why these skills matter

Real access routes include knots, damaged-section isolation, separate drops and changes in work position.

WHEN / WHEN NOT

Use only with a planned destination

Verify the target ropes / anchors and manage swing, spacing and rescue implications.

WHAT GOES WRONG

Committing too early

Unloading the old system before the new one is complete, attaching both devices to one rope, or creating excessive slack.

EVALUATOR'S EYE

Evaluator focus

Deliberate sequence, correct system identification and the ability to pause safely mid-transfer.

TRAINEE Q&A

“Can I transfer faster by moving both devices together?”

Speed is secondary. Maintain a clear support / protection strategy through each individual step.

Knot-Pass Exercise — Pass a Knot on Descent



SEQUENCE — NEVER FEWER THAN TWO ATTACHMENTS

1. **Stop just above the knot**
   Lock off the descender; backup kept high

2. **Clip a short cowstail into the knot loop**
   Safety connection, gate locked

3. **Fit the ascender above the descender, stand up**
   Weight onto the ascender, then remove the descender

4. **Refit the descender below the knot**
   Lock it, load it, then remove the ascender and cowstail

5. **Pass the backup separately**
   Only if the knot is on the backup rope too — stay clipped while moving it

Check: the new device is loaded and locked before the old one comes off.

Rope-to-Rope Transfer — Move Across Without Losing Two Points



SEQUENCE

1. **Stop level with the new rope set**
   Descender locked on the current working rope

2. **Connect a cowstail to the new ropes**
   Clip the anchor knot or a rope grab

3. **Fit the backup on the new backup rope**
   Above your attachment point

4. **Fit the descender on the new working rope**
   Lock it, then take your weight onto it

5. **Remove devices from the old ropes**
   One at a time, after the new system holds you

Never unclip the old system until the new working rope and backup are both on and loaded.

CLASSROOM → PRACTICAL

Day 2 Practical Training Plan

- Fall-protection setup discussion at the structure
- Deviation progression: demo → coached → independent
- Re-belay / re-anchor progression
- Passing knots
- Rope-to-rope transfers
- Combined maneuver circuit with evaluator-style observation



Photo: Archer7319 · public domain · Wikimedia Commons

SKYTECH ACADEMY

ROPE ACCESS TRAINING CENTER

OILFIELD | INDUSTRY | SAFETY | EXPERTISE

DAY 3

ANCHORS, LOAD PHYSICS, RIGGING & RESCUE

Build systems that can support work — and recover a casualty safely.



SOLID ANCHORS
TRUST YOUR CONNECTIONS.

LOAD PHYSICS
UNDERSTAND THE FORCES.

RIGGING SYSTEMS
BUILD STRONG, EFFICIENT SETUPS.

RESCUE READY
RECOVER CASUALTIES SAFELY. EVERY TIME.

ANCHOR SELECTION | LOAD PATH & FORCE ANALYSIS | RIGGING TECHNIQUES | SYSTEMS & REDUNDANCY | RESCUE PLANNING | CASUALTY RECOVERY

PLAN STRONG.
RIG RIGHT.
SAVE LIVES.

1–2 Hour Classroom — Anchors, Load Physics, Rigging & Rescue

01 Anchor selection & independence

DISCUSS Verified anchor — or just convenient?

02 Anchor angles & load sharing

DISCUSS What does each leg see at this angle?

03 MBS, WLL / SWL & equipment limits

DISCUSS Where is the working limit written?

04 Lowering & hauling systems

DISCUSS Lower or haul — and why?

05 Releasable rigging & edge management

DISCUSS Can this rig be lowered under load?

06 Rescue fundamentals & Level 1 progression

DISCUSS Is the rescuer safe first?

07 Level 2 rescue through obstacles

DISCUSS What is the plan for every obstacle?

INSTRUCTOR NOTE Seven topics today — keep each tight and lean on demonstrations. Load physics must be understood before rigging, and rigging before rescue.

Anchor Selection & Independence

WHAT / HOW

Verify the structure and load path

Use suitable anchorages for the intended direction and system. Maintain independence and avoid common-mode failure.

WHY

Why “strong-looking” is not enough

Unknown structure, corrosion, poor load direction or shared failure points can defeat otherwise strong equipment.

WHEN / WHEN NOT

Before rigging

Confirm structural suitability through the approved process and protect ropes from edges, heat, chemicals and movement.

WHAT GOES WRONG

Convenience anchoring

Handrails, pipework, cable tray or unverified steel selected simply because they are nearby.

EVALUATOR'S EYE

Evaluator focus

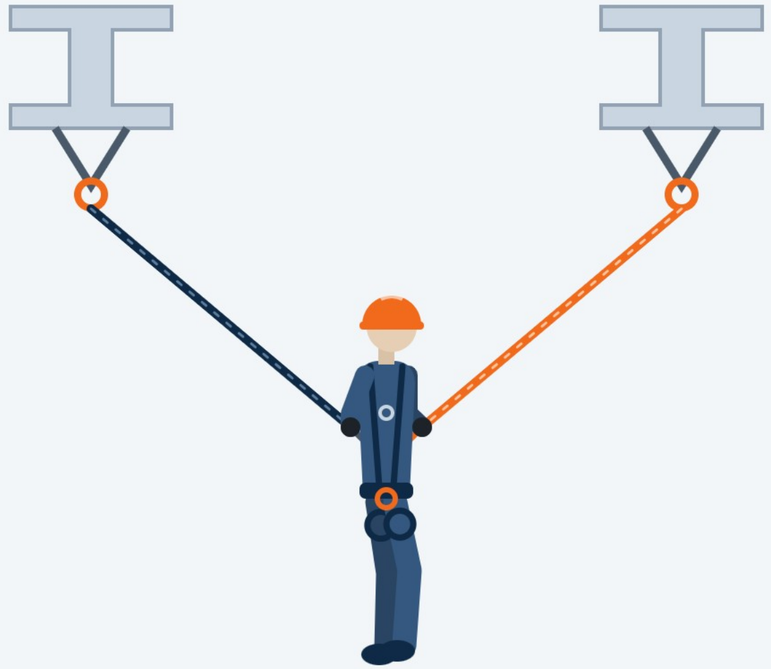
Anchor choice, direction of load, independence, edge protection and rescue implications.

TRAINEE Q&A

“Can I use this beam / rail?”

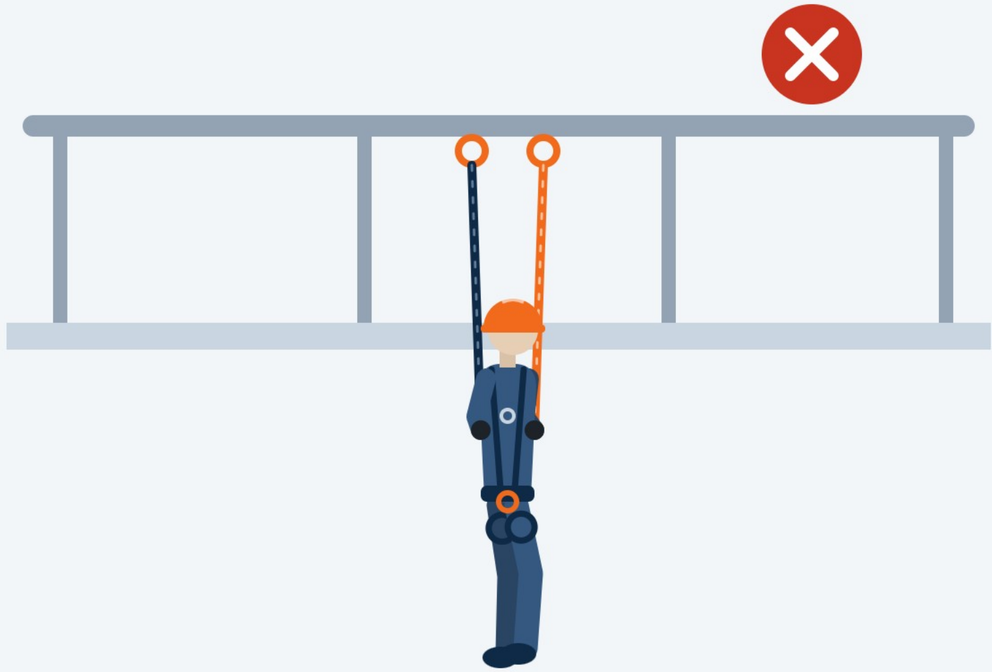
Only after it is verified as suitable under the applicable engineering / site process — not by appearance alone.

Anchor Independence — One Failure Must Not Drop Both Ropes



INDEPENDENT, VERIFIED ✓

Two separate structural members
One anchor failing cannot take the other with it



SHARED, UNVERIFIED ✗

Both ropes on one handrail
Handrail, pipework, cable tray: one failure drops both

THE TEST

Is the structure verified, and can one anchor fail without affecting the other?

THE TRAP

Handrails, pipework or cable tray chosen because they are nearby.

Structural Steel Anchors — Know the Load Case

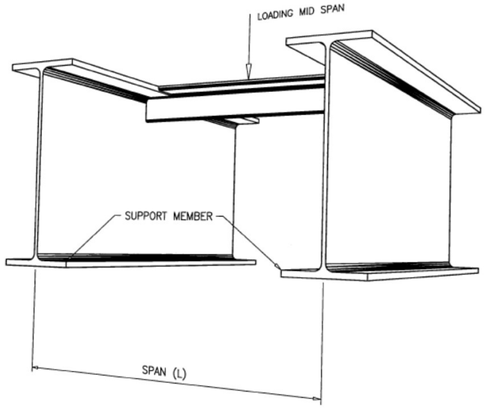


FIG 1 · MIDSPAN

Load hangs between two supports

Span = distance between the supports

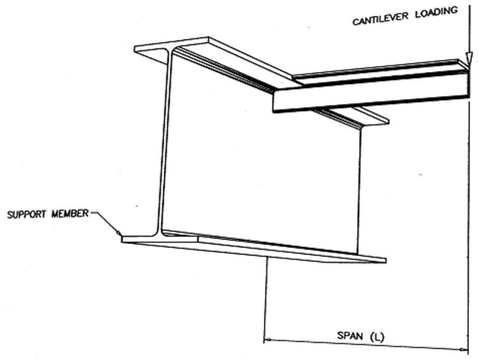


FIG 2 · CANTILEVER

Load at the free end of an overhang

Span = overhang past the support

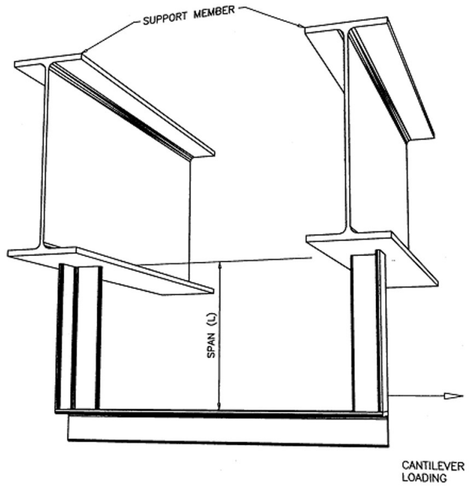


FIG 3 · CANTILEVER (SIDE PULL)

Load pulls sideways on a hanging member

Span = drop below the support

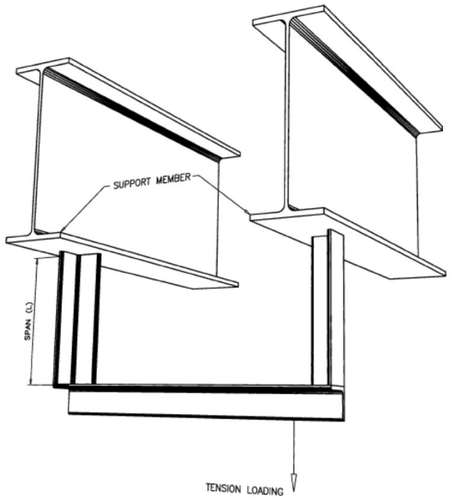


FIG 4 · TENSION

Load pulls straight down the member

Span = drop below the support

Same member, different answer. A beam that holds in tension can fail as a cantilever. Identify the load case and the span before you pick the anchor.

Is This Steel Member a Suitable Anchor?

	MIDSPAN			CANTILEVER			TENSION		
	2 ft	5 ft	10 ft	2 ft	5 ft	10 ft	2 ft	5 ft	10 ft
I-BEAM (W)									
W4	YES	YES	NO	NO	NO	NO	YES	YES	YES
W6	YES	YES	NO	NO	NO	NO	YES	YES	YES
W8	YES	YES	YES	YES	NO	NO	YES	YES	YES
W10	YES	YES	YES	YES	NO	NO	YES	YES	YES
W12	YES	YES	YES	YES	YES	NO	YES	YES	YES
SQUARE TUBE (TS)									
TS 2×2	NO	NO	NO	NO	NO	NO	YES	YES	YES
TS 3×3	YES	NO	NO	NO	NO	NO	YES	YES	YES
TS 4×4	YES	NO	NO	NO	NO	NO	YES	YES	YES
TS 5×5	YES	YES	NO	NO	NO	NO	YES	YES	YES
TS 6×6	YES	YES	YES	YES	NO	NO	YES	YES	YES

	MIDSPAN			CANTILEVER			TENSION		
	2 ft	5 ft	10 ft	2 ft	5 ft	10 ft	2 ft	5 ft	10 ft
PIPE (SCHD 40)									
2″	NO	NO	NO	NO	NO	NO	NO	NO	NO
3″	YES	NO	NO	NO	NO	NO	YES	YES	YES
4″	YES	NO	NO	NO	NO	NO	YES	YES	YES
5″	YES	YES	NO	NO	NO	NO	YES	YES	YES
6″	YES	YES	YES	YES	NO	NO	YES	YES	YES
ANGLE (L)									
L 3×3	NO	NO	NO	NO	NO	NO	NO	NO	NO
L 4×4	NO	NO	NO	NO	NO	NO	YES	YES	YES
L 5×5	YES	NO	NO	NO	NO	NO	YES	YES	YES
L 6×6	YES	NO	NO	NO	NO	NO	YES	YES	YES
CHANNEL (C)									
C4	YES	NO	NO	NO	NO	NO	YES	YES	YES
C6	YES	YES	NO	NO	NO	NO	YES	YES	YES
C8	YES	YES	—	YES	NO	—	YES	YES	—
C10	YES	YES	YES	YES	NO	NO	YES	YES	YES

READ IT LIKE THIS

Tension is almost always YES. Cantilever is almost always NO. Longer span = weaker anchor.

Acceptable per fall-protection standard for the loading shown (figures on the previous slide). Blank = not listed. Planning aid only — a qualified person approves every anchor.

Anchor Angles & Load Sharing

WHAT / HOW

Geometry changes force

Explain how the included angle affects leg tension and why increasingly wide bridles create high forces.

WHY

Why this matters

A multi-anchor system can overload components if geometry is poor — even when the suspended load itself seems modest.

WHEN / WHEN NOT

Keep geometry conservative

Use manufacturer / engineering guidance for exact limits and avoid unnecessarily wide included angles.

WHAT GOES WRONG

False “equalization” assumptions

Unequal leg lengths, stretch, direction changes and movement can prevent true equal load sharing.

EVALUATOR'S EYE

Evaluator focus

Candidate recognizes poor geometry and can explain why flattening a bridle increases leg force.

TRAINEE Q&A

“Is 120° always safe?”

No angle is automatically safe. The complete engineered / approved system and component limits control.

Included Angles — the Wide V Multiplies the Load

ANGLE	FORCE / ANCHOR	STATUS
0°	50% of load	Ideal
45°	54%	Good
90°	71%	Limit of preferred
120°	100% — no sharing	Critical
150°	193%	Danger
160°	288%	Extreme danger

WORKED EXAMPLE — DO THE MATH

ANGLE	LOAD	FORCE / LEG
90°	220 lb	≈ 156 lbf
120°	180 lb	180 lbf
160°	220 lb	≈ 634 lbf

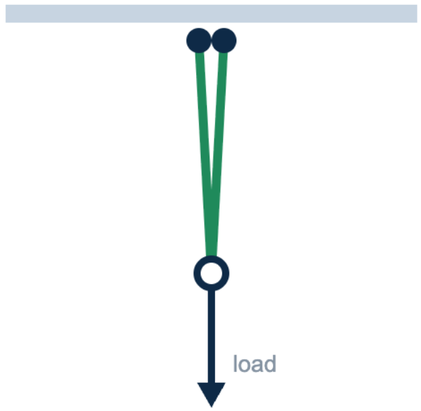
At 160° each anchor sees **nearly 3× the entire load** — geometry alone made it dangerous.

THE RULES

Keep the included angle at 90° or less where practical.

Wider than a right angle? Re-rig with longer slings or a higher master point.

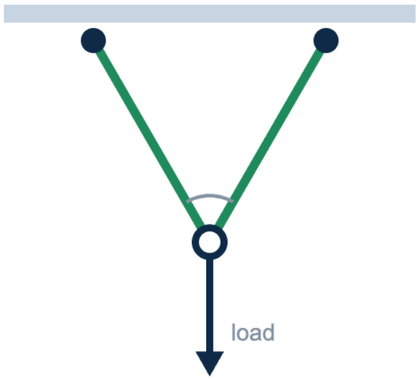
Anchor Angles — Watch the Load Grow



0°

50% per leg

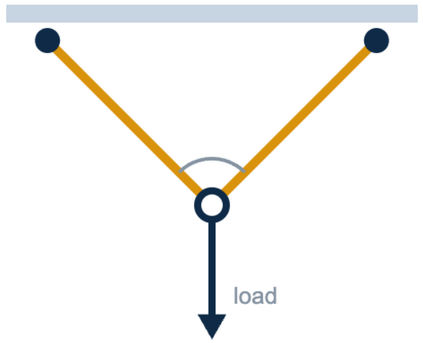
Ideal



60°

58% per leg

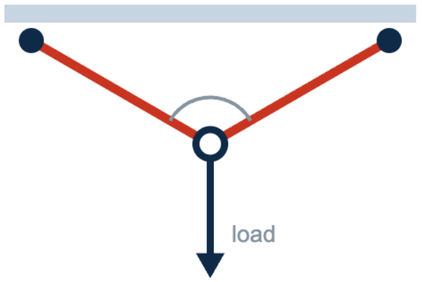
Good



90°

71% per leg

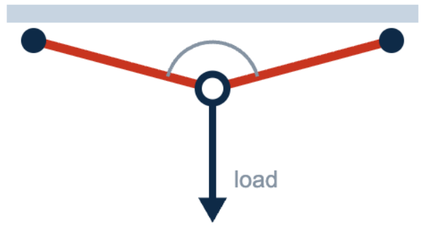
Preferred limit



120°

100% per leg

No sharing



150°

193% per leg

Danger

THE RULE

Force per leg = load ÷ (2 × cos(½ × angle)). Keep the included angle at 90° or less. At 120° each anchor holds the full load; past that, each anchor holds more than the load itself. Too wide? Re-rig with longer slings or a higher master point.

Y-Hang — Share the Load Between Two Anchors



Illustration: vRigger

BUILD IT

1. **Two independent anchors, each rated**
   Either one alone must be able to hold the full load

2. **Tie into the first anchor**
   Figure 8 on a bight, clipped with a locked connector

3. **Run the second leg to the other anchor**
   Clip the bight; set leg length before you tie off

4. **Tie the Y-hang knot at the center**
   Both legs share the load; knot dressed and set

5. **Check the angle between the legs**
   Aim for 90° or less — never wider than 120°

At 120° each anchor carries 100% of the load. Wider than that, each anchor carries more than the load itself.

MBS, WLL / SWL & Equipment Limits

WHAT / HOW

Read the right value

Differentiate minimum breaking strength from allowable working limits and manufacturer / system restrictions.

WHY

Why labels can be misread

A breaking-strength number is not automatically an allowable operating load and does not replace system design.

WHEN / WHEN NOT

Use approved limits

Follow manufacturer, applicable standard, engineering and organizational procedures.

WHAT GOES WRONG

Universal division rules

Do not teach “MBS ÷ 5” or “÷ 10” as a universal rope-access WLL formula.

EVALUATOR'S EYE

Evaluator focus

Candidate knows when a rating can be used and when engineering / manufacturer data is required.

TRAINEE Q&A

“Can I calculate WLL from MBS myself?”

Not as a universal shortcut. Use the governing approved data / method for that equipment and system.

WLL vs. MBS — Breaking Point vs. Working Limit

MBS

Minimum Breaking Strength — the failure point. Reference only.

WLL

Working Load Limit — the safe load you rig to.

HOW THE EXAMPLE IS CALCULATED

Example WLL = MBS ÷ design factor. This table uses metal **÷5** and textile / mechanical **÷10**.

On the job, rig to the WLL the manufacturer states — factors vary by product.

WLL assumes a static, in-line load. Shock loading and bad angles eat the margin fast.

EQUIPMENT	MBS	IN LBF ÷ FACTOR	EXAMPLE WLL
Carabiner	27 kN	**6,070 ÷ 5**	**1,214 lbf**
Rope (with knot)	6,000 lbf	**4,200* ÷ 10**	**420 lbf**
Rigging plate	36 kN	**8,093 ÷ 5**	**1,619 lbf**
Pulley (body)	23 kN	**5,170 ÷ 5**	**1,034 lbf**
I'D descender	14 kN	**3,147 ÷ 10**	**315 lbf**

1 kN = 224.8 lbf · * 6,000 lbf × 0.70 for knot loss (example)

System WLL = 315 lbf — the weakest link sets it.

The Weakest Link Sets the System Rating



THE RULE

System WLL = the lowest WLL of any component in the load path.

THE TAKEAWAY

Upgrading a strong component changes nothing. Only the weakest one raises the rating.

Lowering & Hauling Systems

WHAT / HOW

Control load movement

Teach lowering devices, mechanical advantage, progress capture, redirects, backups, edge management and communication.

WHY

Why system efficiency matters

Friction, redirects and poor geometry change actual effort and forces — a theoretical haul ratio is not the real-world result.

WHEN / WHEN NOT

Choose the simplest safe system

Match the system to casualty / load, distance, obstacles, available rescuers and anchors.

WHAT GOES WRONG

Complexity without control

Too many components, poor progress capture, crossed ropes, overloaded redirects and unclear commands.

EVALUATOR'S EYE

Evaluator focus

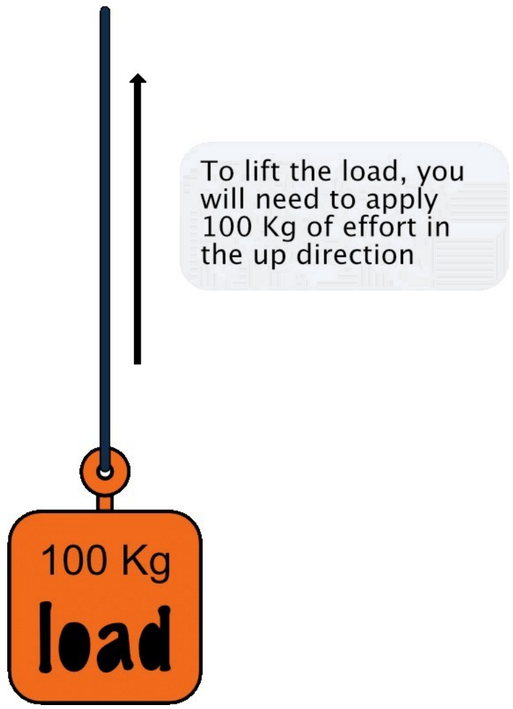
Correct rigging, backup, load control, communication and the ability to stop / hold safely.

TRAINEE Q&A

“Is a 4:1 always better than a 3:1?”

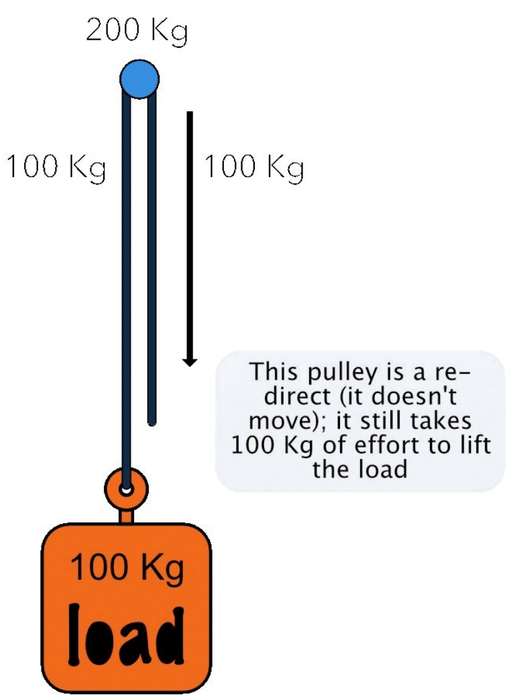
No. More advantage adds rope travel, equipment and friction. Use what the scenario requires.

Mechanical Advantage — Count the Lines



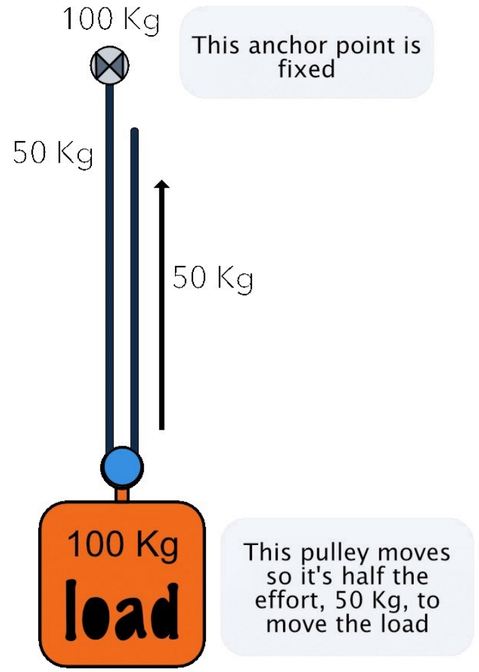
1:1 · DIRECT LIFT

Effort = load



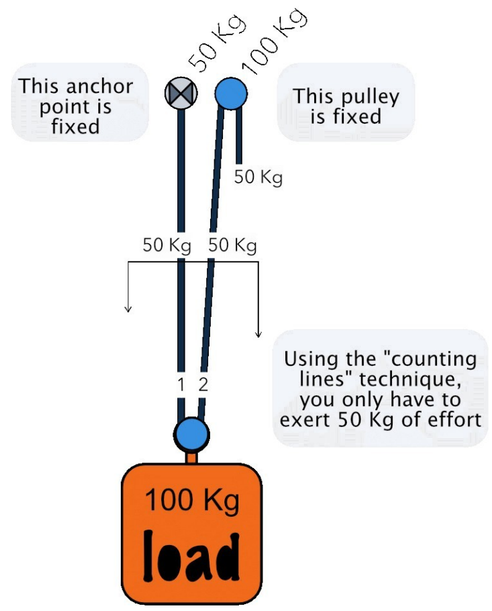
1:1 · REDIRECT

A fixed pulley only changes direction



2:1 · MOVING PULLEY

Pulley rides on the load

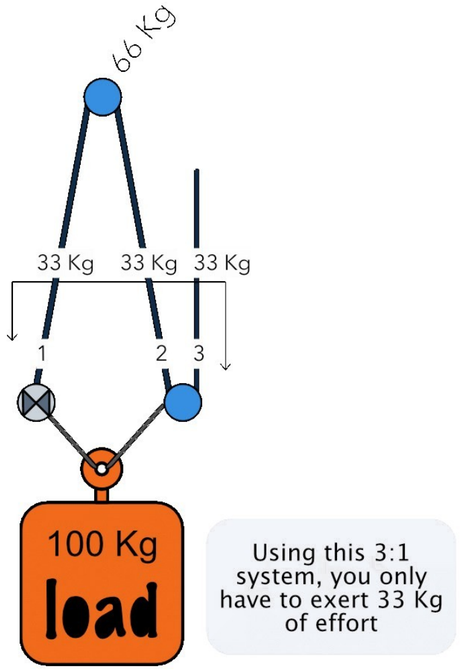


2:1 · COUNTING LINES

2 lines hold the load → 50 kg

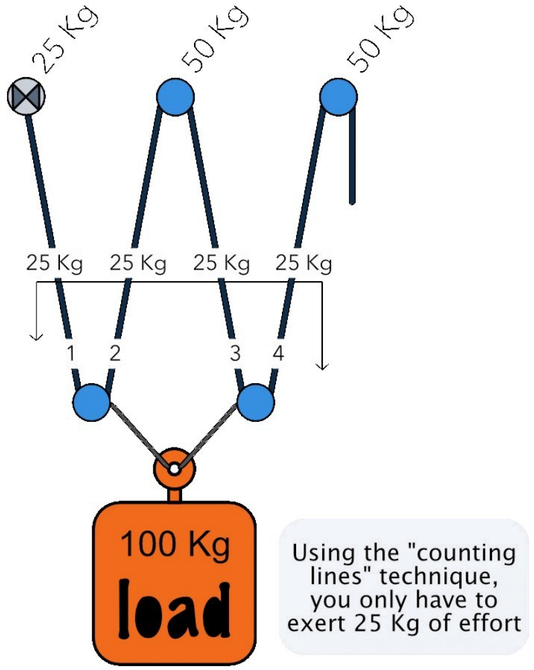
Count the lines that hold the load. Two lines on a moving pulley = half the effort.

Simple Systems — 3:1, 4:1, 5:1



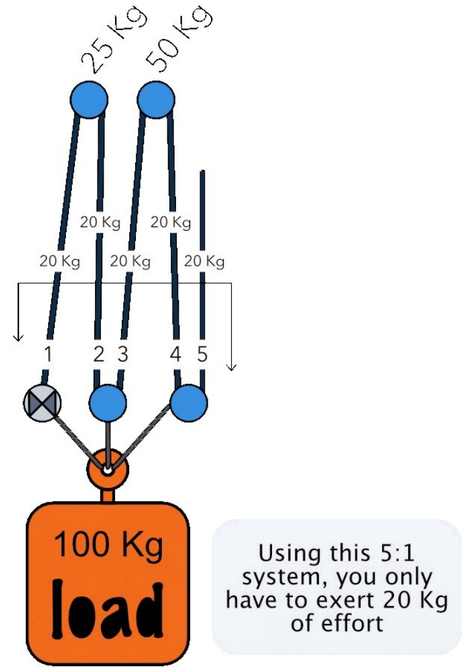
3:1

3 lines → 33 kg effort



4:1

4 lines → 25 kg effort

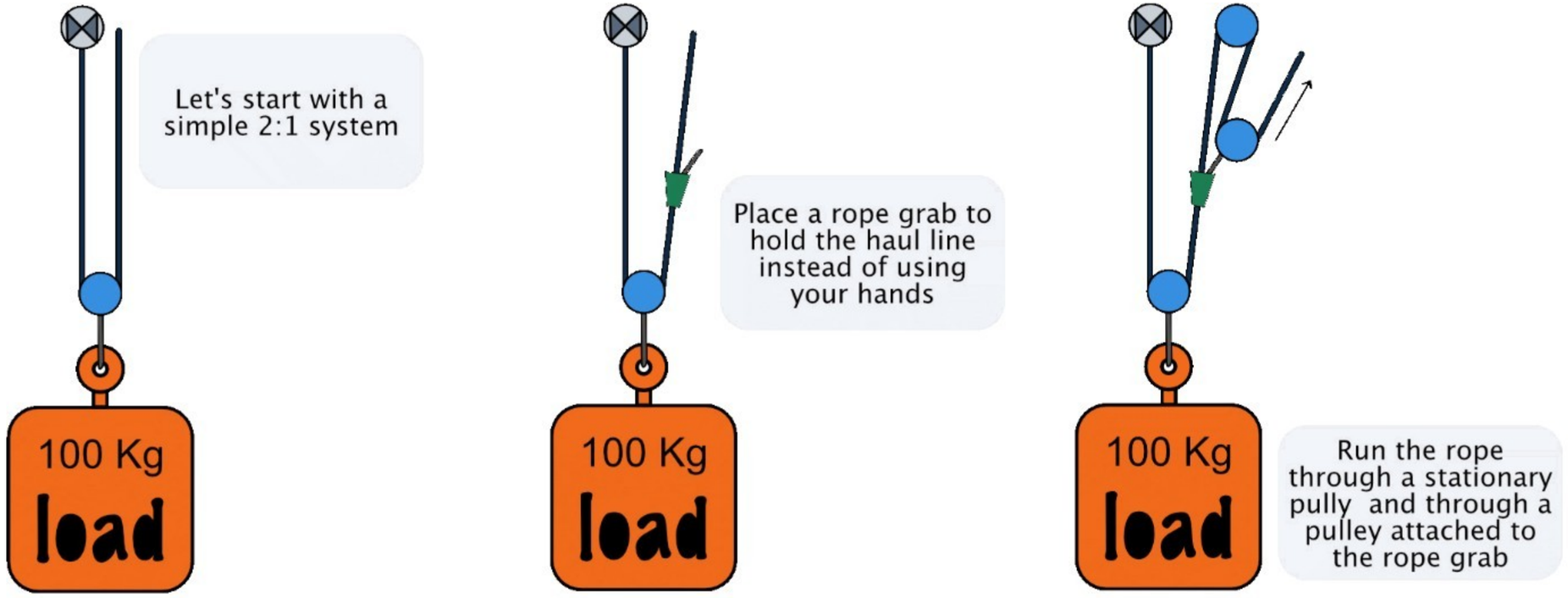


5:1

5 lines → 20 kg effort

Effort = load ÷ lines. Every pulley adds friction, so real effort is higher than the math.

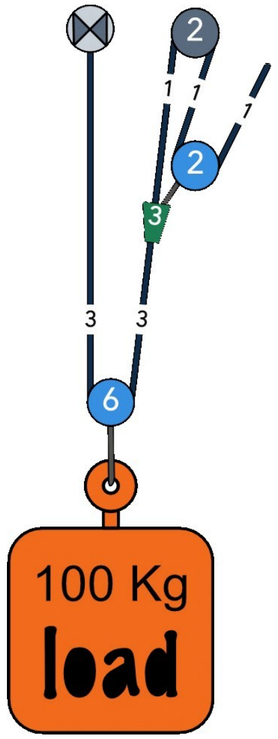
Building a Compound System — Step by Step



START WITH 2:1 · ADD A ROPE GRAB · ADD A SECOND SYSTEM ON THE GRAB

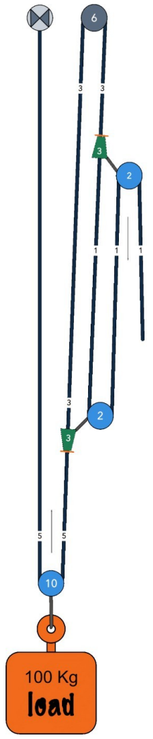
A rope grab on the haul line becomes the “load” for a second system — the ratios multiply.

Compound Systems — Multiply the Ratios



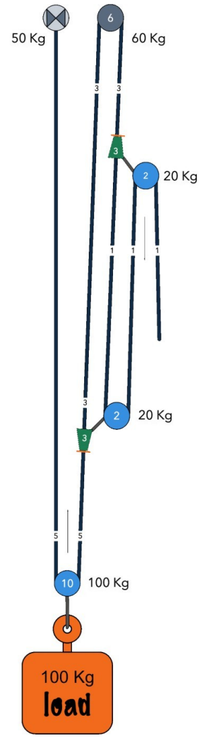
3:1 ON 2:1 = 6:1

1 unit in → 6 at the load



10:1 · T-METHOD

1 unit in → 10 at the load



10:1 · IN KG

10 kg effort → 100 kg load

3:1 on a 2:1 = 6:1. 5:1 on a 2:1 = 10:1. Higher ratio = less effort, more rope to pull and more friction.

Releasable Rigging & Edge Management

WHAT / HOW

Rig with rescue in mind

Where appropriate, build systems that allow controlled lowering / transfer under load and protect ropes at edges.

WHY

Why pre-planning saves time

A non-releasable system may force rescuers to build complex alternatives while a casualty remains suspended.

WHEN / WHEN NOT

Use where the risk assessment supports it

Consider work position, access, rescue direction, anchor capacity and edge conditions.

WHAT GOES WRONG

Unprotected edges / locked systems

Abrasion, cutting, difficult casualty transfer and inability to lower a loaded system.

EVALUATOR'S EYE

Evaluator focus

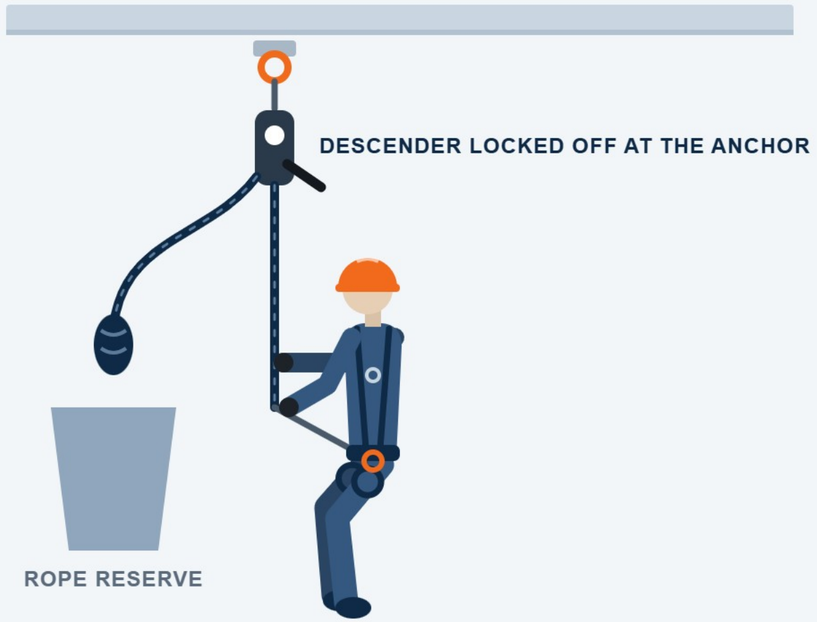
Rescue compatibility is considered during initial rigging — not added only after an incident.

TRAINEE Q&A

“Should every rig be releasable?”

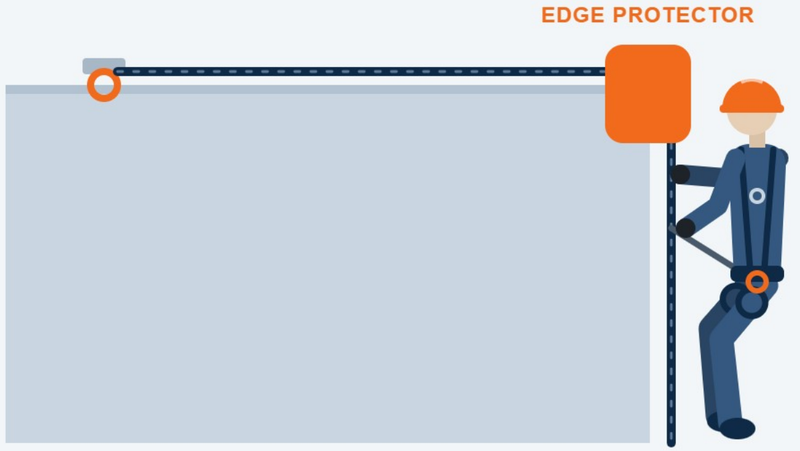
Use the method the approved plan and conditions require; the key is that a workable rescue strategy exists.

Rig for Rescue — Releasable Anchors and Protected Edges



RELEASABLE ANCHOR

Unlock and lower the loaded rope:
no cutting, no complex rescue build



EDGE MANAGEMENT

Protect every point where the rope touches an edge
No bare rope over a sharp edge

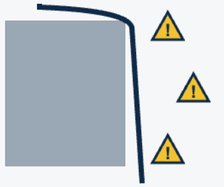
THE PLAN

Rescue compatibility is built in at the start, not added after an incident.

THE EDGE

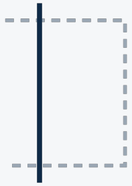
A protector at every edge contact; reroute or deviate the rope where you can.

Protecting the Rope — Work Down the List



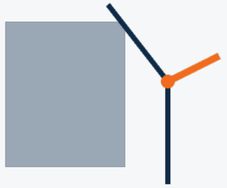
1 IDENTIFY THE HAZARD

Walk the whole rope path. Look for sharp edges, hot surfaces, abrasion, chemicals, loose tools, wind, entanglement and entrapment.



2 REMOVE IT

Grind or cover the sharp edge, shut off the heat source, clear anything the rope could snag on.



3 AVOID IT

Move the rope away from the hazard: alternative anchor, Y-hang, deviation or re-anchor.



4 PROTECT AGAINST IT

Edge protector at edges, rope protector on other surfaces. Each rope gets its own protection.



5 VERIFY

Check the protection is right and secure. If it is not, do not go on rope.

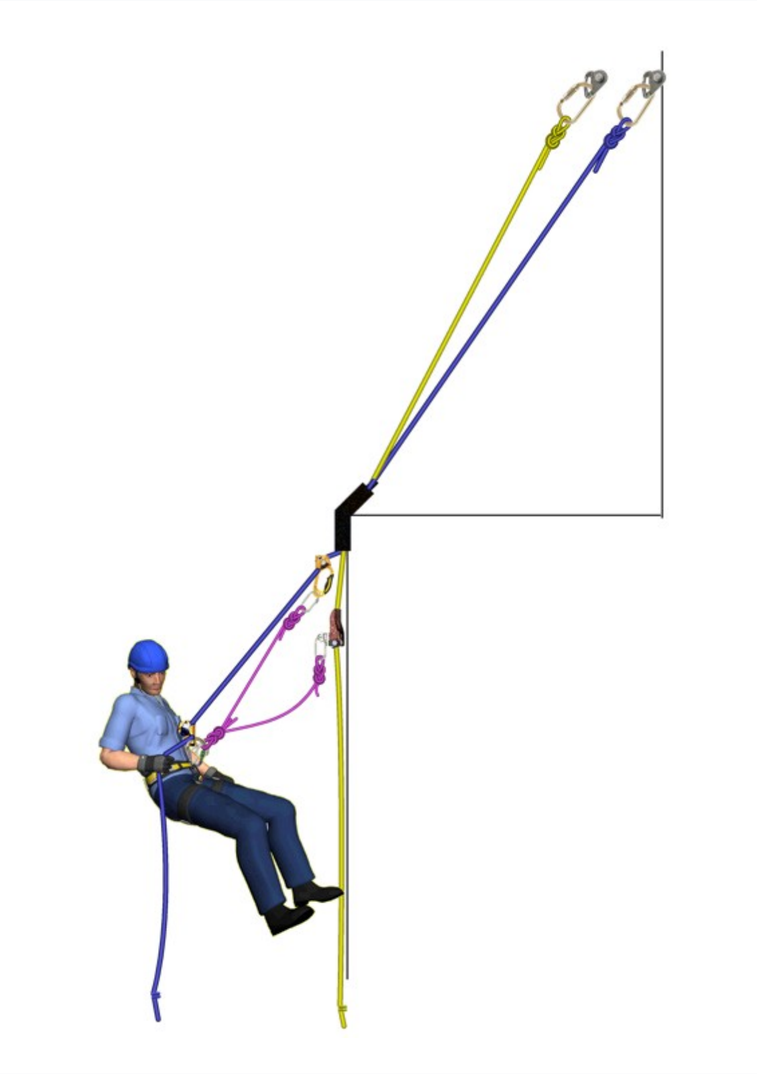
WORK DOWN THE LIST

Try each step in order. Only move to the next step when the one above cannot remove the risk.

SAFETY-CRITICAL

Fabric or canvas rope protectors guard against abrasion only. A sharp edge needs rigid edge protection — or re-rigging so the rope never touches it.

Going Over the Edge — Protect the Rope at the Lip



SET-UP AND MOVE

1. **Anchor back from the edge**
   Each rope on its own anchor, rigged before you approach

2. **Protector at every contact point**
   Secured so it can’t slide off the lip

3. **Backup rope protected too**
   Both ropes run over protection, not bare edge

4. **Sit back, feet low, then step over**
   Weight in the harness before the feet leave the top

5. **Check rope position below the lip**
   Rope runs clear; no new rub point

No bare rope over an edge. If you can’t protect it, re-route or deviate the rope.

Rescue Fundamentals & Level 1 Progression

WHAT / HOW

Make safe → access → stabilize → transfer → move → handover

SPRAT L1 rescue: reach a casualty in ascent mode on an adjacent rope set, convert them to descent, move both onto one system and descend.

WHY

Why rescue starts before the casualty

The rescuer must not become a second casualty. Hazard control, redundancy and command come first.

WHEN / WHEN NOT

Simple, predictable scenarios first

Build competence before adding deviations, re-belays, transfers or complex hauling.

WHAT GOES WRONG

Rushing the casualty connection

Poor rescuer attachment, uncontrolled transfer, dropped backup, bad communication or uncontrolled descent.

EVALUATOR'S EYE

Evaluator focus

Rescuer safety, casualty security, controlled transfer and correct final descent / landing.

TRAINEE Q&A

“Do I rescue as fast as possible?”

Work promptly, but never sacrifice rescuer / casualty safety or system control for speed.

Rescue From Ascent Mode — Reach, Connect, Convert, Descend



SEQUENCE

1. **Ascend to the casualty**
   Stop just below them; your own system stays complete

2. **Connect the casualty to your harness**
   Short connection — keep them close

3. **Take the casualty’s weight**
   Onto your descender, using a mini-haul if needed

4. **Remove the casualty’s ascenders**
   Their backup stays on until they are on your system

5. **Descend together under control**
   Brake hand on; backup managed for both

Rescuer safety first. Check your own system before you touch theirs.

Level 2 & 3 Rescue Through Obstacles

WHAT / HOW

Move a casualty through system complexity

L2: same-rope rescue with a rope-to-rope transfer, rescue from fall-arrest lanyards and a remote lowering system. L3: rescue from two maneuvers.

WHY

Why obstacles change rescue

Two-person loads, limited mobility and device congestion make ordinary maneuvers significantly more complex.

WHEN / WHEN NOT

Plan the entire route

Identify where load transfers, redirects and rescuer repositioning will occur before committing.

WHAT GOES WRONG

Solving one obstacle at a time without a plan

Can trap the rescuer / casualty, overload a system or create an impossible next step.

EVALUATOR'S EYE

Evaluator focus

Planning, two-person load control, redundancy, obstacle sequence and efficient communication.

TRAINEE Q&A

“What is the best rescue method?”

The simplest approved method that safely fits the casualty, geometry, hazards, equipment and team capability.

Rescue From Descent Mode — Take Over the Casualty’s Load



SEQUENCE

1. **Approach on your own system**
   Ascend or descend to just below the casualty

2. **Connect the casualty to you**
   Short connection to your attachment point

3. **Transfer their weight to your descender**
   Release their descender only after yours is loaded

4. **Manage both backups**
   Casualty backup removed last

5. **Descend together**
   Slow and controlled; hand over at the ground

The key step: a controlled transfer of the casualty’s weight onto the rescuer’s system.

CLASSROOM → PRACTICAL

Day 3 Practical Training Plan

- Anchor identification and rigging exercises
- Angle / load demonstrations and system critique
- Lowering and hauling systems
- Releasable rigging / edge-management stations
- Level 1 rescue progression
- Level 2 obstacle-rescue progression

SPRAT TASKS PRACTICED TODAY

- **L1** Build a two-rope system
- **L1** Raise & lower from the ground
- **L1** Pick-off rescue, adjacent ropes
- **L2** Remote rescue (lowering) system
- **L2** Rescue from fall-arrest lanyards
- **L2** Same-rope rescue + transfer

SKYTECH ACADEMY
ROPE ACCESS TRAINING CENTER

OILFIELD | INDUSTRY | SAFETY | EXPERTISE

DAY 4

ADVANCED SKILLS, RESCUE SCENARIOS & CONSOLIDATION

Develop Level 2/3 judgment, team rescue management and evaluation readiness.



TEAM RESCUE MANAGEMENT
LEAD. COORDINATE. SAVE TOGETHER.

COMPLEX RESCUE SCENARIOS
PRACTICE. SOLVE. PERFORM.

EVALUATION READINESS
MEET STANDARDS. PROVE COMPETENCE.

LEVEL 2/3 JUDGMENT
MAKE THE RIGHT CALL. EVERY TIME.

ADVANCED ROPE CONTROL
Improve efficiency and precision on rope.

COMPLEX RIGGING APPLICATIONS
Build stronger, smarter rigging systems.

TEAM RESCUE SCENARIOS
Realistic scenarios. Real teamwork.

RESCUE MANAGEMENT FLOW & COMMUNICATION
Clear communication. Effective execution.

FINAL EVALUATION & SKILLS CONSOLIDATION
Consolidate skills. Show you're ready.

TRAIN TOGETHER.
THINK AHEAD.
BE READY.
SAVE LIVES.

1–2 Hour Classroom — Advanced Skills, Rescue Scenarios & Consolidation

01 **Advanced rigging & complex re-anchors**

DISCUSS Is this the simplest system that works?

02 **Horizontal / tensioned systems & aid climbing**

DISCUSS What does tension do to the anchors?

03 **Complex casualty movement**

DISCUSS Who commands the load?

04 **Level 3 rescue management**

DISCUSS What is Plan B?

05 **SPRAT level distinctions (L1 · L2 · L3)**

DISCUSS What does each SPRAT level add?

06 **Mock evaluation & targeted remediation**

DISCUSS What did the mock reveal?

INSTRUCTOR NOTE Today is where judgment matters more than technique. Emphasize simplicity, command and rescue foresight over adding components.

Advanced Rigging & Complex Re-anchors

WHAT / HOW

Design the route before loading it

Plan anchor layout, rope path, redundancy, rescue compatibility and transitions for more complex access.

WHY

Why planning dominates advanced work

Complex systems multiply the interaction between anchors, ropes, devices, edges and people.

WHEN / WHEN NOT

Only within competence and approved design

Escalate engineered / high-load systems and unusual structures to the appropriate qualified authority.

WHAT GOES WRONG

Building complexity for its own sake

Unnecessary redirects, inaccessible knots, poor rescue paths and unclear load distribution.

EVALUATOR'S EYE

Evaluator focus

System simplicity, clean rigging, independence, load path and rescue foresight.

TRAINEE Q&A

“What makes rigging ‘advanced’?”

Not the number of components — the complexity of geometry, loading, access, rescue and supervision decisions.

Horizontal Aid Climbing — Two Connections at All Times



SEQUENCE

1. **Clip the lead cowstail to the next anchor**
   Gate locked before any weight goes on

2. **Clip the étrier and step up**
   Feet in the étrier, weight off the trailing cowstail

3. **Clip the second cowstail forward**
   Now on two anchors again

4. **Remove the trailing connection**
   Only once both forward connections are loaded

5. **Repeat, anchor by anchor**
   SPRAT L1: 3 m horizontal · L2: 3 m vertical or angled

Swing fall ≤ 2 ft. Plan anchor spacing so a slip cannot swing you into the structure.

Horizontal / Tensioned Systems & Aid Climbing

WHAT / HOW

Maintain continuous protection while moving laterally

SPRAT progression: L1 aid-climbs 3 m horizontally, L2 aid-climbs 3 m vertically or at an angle, L3 builds and uses a tensioned rope system.

WHY

Why lateral movement changes risk

Anchors may see different directions and forces, and the worker can experience larger swings or difficult rescue access.

WHEN / WHEN NOT

Use approved system design

Tensioned systems require careful force management — do not improvise high-tension rigging.

WHAT GOES WRONG

Over-tensioning and poor backup

High anchor forces, insufficient redundancy, long fall / swing paths and difficult casualty recovery.

EVALUATOR'S EYE

Evaluator focus

Controlled progression, connection sequence, system-force awareness and a rescue plan.

TRAINEE Q&A

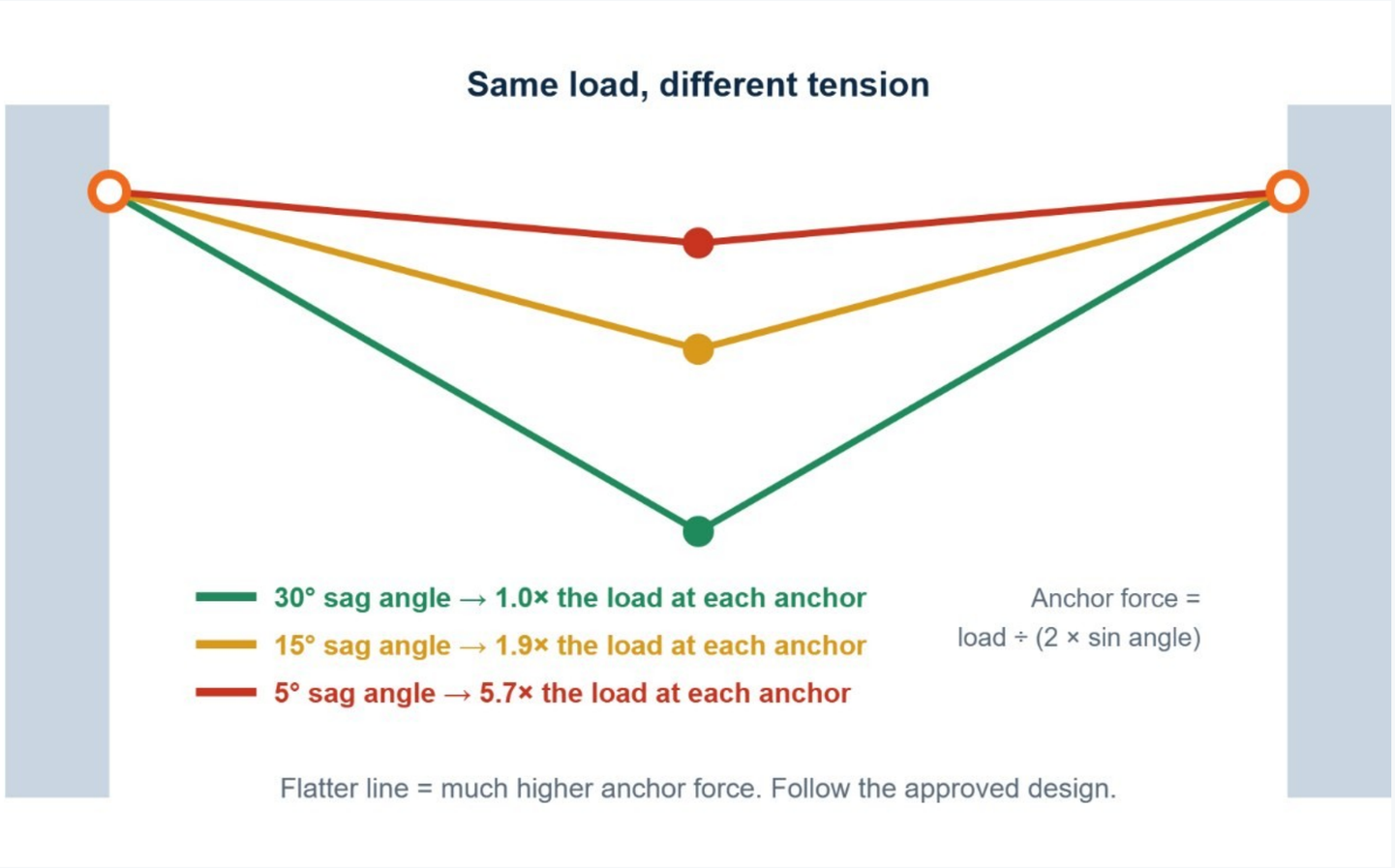
“Can I pull the line tighter to reduce sag?”

Increasing tension can greatly increase anchor / system forces. Follow the approved design and limits.

Tensioned Lines — Less Sag, Much Higher Anchor Force



Photo: Masa Sakano · CC BY-SA 2.0 · Wikimedia Commons



THE PHYSICS

Anchor force = load ÷ (2 × sin of the sag angle).

THE RULE

Follow the approved design; never over-tension a line to reduce sag.

Tensioned Lines — Anchor Force by Sag Angle

SAG ANGLE	MULTIPLIER	1 PERSON · 220 LB	RESCUE · 440 LB
30°	1.00 ×	220 lbf	440 lbf
15°	1.93 ×	425 lbf	850 lbf
10°	2.88 ×	633 lbf	1,267 lbf
5°	5.74 ×	1,262 lbf	2,524 lbf

Force at EACH anchor = load ÷ (2 × sin sag angle). Example loads; real loads are dynamic.

Halve the sag angle and the anchor force roughly doubles. A tight line can load each anchor with several times the rescue load.

DO THE MATH

Two-person rescue load, 10° sag

sin 10°	= 0.174
2 × 0.174	= 0.347
440 lb ÷ 0.347	= 1,267 lbf

at each anchor — nearly 3× the load.

THE RULE

Rig to the approved design and its anchor ratings. Never over-tension a line to reduce sag.

Tensioned-Line Traverse — Rig It, Ride It, Check It



SET-UP

1. **Two lines, each with its own anchors**
   A failure of one line must not drop the technician

2. **Anchors rated for tension forces**
   Use the sag-angle math from the previous slide

3. **Connect to both lines**
   Pulleys or connectors on each line, cowstails locked

4. **Control the travel**
   Haul or tag line to move; never free-wheel

5. **Check sag under load**
   Confirm the angle matches the approved design

Never over-tension a line to reduce sag — anchor force rises fast.

Complex Casualty Movement

WHAT / HOW

Control two-person systems through obstacles

Coordinate rescuer position, casualty orientation, load transfer, hauling / lowering and edge / obstacle passage.

WHY

Why small errors become large

Casualty weight, device congestion and limited mobility reduce the margin for improvisation.

WHEN / WHEN NOT

Use staged checkpoints

Pause in stable positions before each major transfer or obstacle.

WHAT GOES WRONG

Unplanned simultaneous actions

Multiple rescuers moving devices or loads without one command can create shock loads or loss of control.

EVALUATOR'S EYE

Evaluator focus

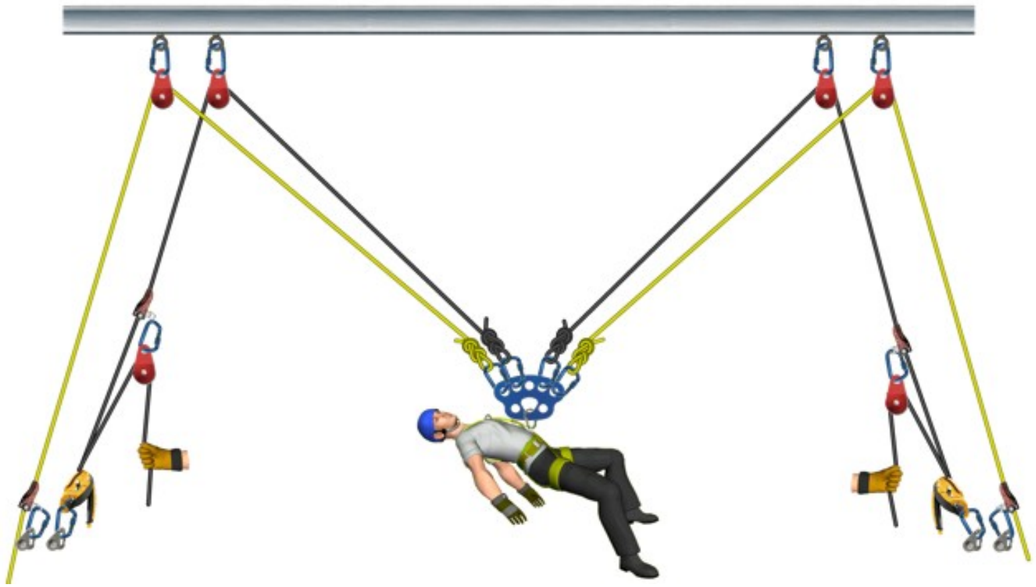
Stable staging, clear commands, controlled load transfer and casualty protection.

TRAINEE Q&A

“Who should move first?”

The rescue plan and command structure decide; only one coordinated plan should control the load.

Cross-Haul — Move the Casualty Away From the Structure



HOW IT WORKS

1. **Two haul systems from separate overhead anchors**
   Each side has its own main line and backup line

2. **Casualty hangs from a rigging plate**
   Lines from both sides meet at the plate above the casualty

3. **Haul one side, lower the other**
   Shifting the balance moves the casualty sideways

4. **Progress capture on every haul line**
   A release of the hand never drops the load

5. **One leader gives every command**
   Both teams move only on the leader’s call

Watch the angles. A flat V between the two sides multiplies anchor loads — the same physics as the Y-hang.

Level 3 Rescue Management

WHAT / HOW

Size up, command, plan, execute, terminate

Assign roles, select Plan A / Plan B, and manage communications, resources, fatigue and the medical interface.

WHY

Why leadership is a technical skill

A correct rope system can still fail operationally if commands, roles or contingency planning are weak.

WHEN / WHEN NOT

Use closed-loop communication

One rescue lead; commands are repeated and confirmed before any load movement.

WHAT GOES WRONG

Everyone solving independently

Conflicting commands, duplicated actions, equipment confusion and no shared fallback plan.

EVALUATOR'S EYE

Evaluator focus

Leadership, hazard control, method selection, communication and adaptation without losing system safety.

TRAINEE Q&A

“Should the Level 3 do every rope task?”

No. The supervisor manages the operation and delegates while maintaining control and oversight.

SPRAT Level Distinctions

WHAT / HOW

Teach the current SPRAT documents — not a blended homemade version

Keep certification levels, experience / upgrade requirements, evaluation methods and terminology tied to the applicable official documents.

WHY

Why the distinction matters

Each SPRAT level carries its own responsibilities, experience requirements and evaluation criteria. Mixing them creates incorrect expectations or eligibility advice.

WHEN / WHEN NOT

Verify before every course cycle

Use current official certification / training requirements with documented revision control.

WHAT GOES WRONG

Outdated hour / month claims

Do not rely on simplified remembered upgrade rules when the official scheme document controls.

EVALUATOR'S EYE

Evaluator focus

Instructor materials match the candidate’s actual SPRAT level.

TRAINEE Q&A

“Can I skip straight to Level 2 or 3?”

No. Each level has its own experience and evaluation requirements under current SPRAT rules.

SPRAT Levels — Who Does What on the Rope

LEVEL 1

Rope access technician

Works under Level 3 supervision

Ascent, descent, basic manoeuvres

Performs a pick-off rescue

LEVEL 2

Lead technician

Rigging and complex manoeuvres

Performs rescues

Works under Level 3 supervision

LEVEL 3

Supervisor

Responsible for the worksite and team

Rescue plan and system checks

Full rescue capability

Experience hours and upgrade rules: always check the current SPRAT certification documents.

THE STRUCTURE

Level 1 and Level 2 technicians work under a Level 3 supervisor.

THE SOURCE

Experience and upgrade requirements come from the current SPRAT documents only.

Mock Evaluation & Targeted Remediation

WHAT / HOW

Simulate independent performance

Run equipment / theory, maneuver, rigging, rescue and scenario stations appropriate to the candidate level.

WHY

Why the mock helps

It reveals whether the trainee can perform without step-by-step instructor prompting.

WHEN / WHEN NOT

Use it as a diagnostic

Identify specific gaps and remediate them safely before final assessment.

WHAT GOES WRONG

Turning the mock into nonstop coaching

If the instructor supplies every next step, the exercise no longer tests independent competence.

EVALUATOR'S EYE

Evaluator focus

Safety, sequence, judgment, self-checking and recovery from minor errors within SPRAT rules.

TRAINEE Q&A

“What should I practice tonight?”

Focus on identified weak areas, review theory, and rest. Fatigue is not a training strategy.

CLASSROOM → PRACTICAL

Day 4 Practical Training Plan

- Advanced rigging / complex re-anchor exercises
- Horizontal / tensioned-system or aid-climbing drills as applicable
- Complex casualty movement
- Team rescue scenario with assigned command roles
- Plan A / Plan B scenario changes
- Mock evaluation + targeted remediation

SPRAT TASKS PRACTICED TODAY

- **L2** Cross-haul (lateral operations)
- **L2** Retrievable two-rope system
- **L2** Vertical aid climbing
- **L3** Tensioned rope system
- **L3** Rig a deviation or re-anchor
- **L3** Rescue from two maneuvers

SKYTECH ACADEMY
ROPE ACCESS TRAINING CENTER

OILFIELD | INDUSTRY | SAFETY | EXPERTISE

DAY 5

ASSESSMENT, DEBRIEF & PROFESSIONAL DEVELOPMENT

Demonstrate independent competence and leave with a clear development path.



COMPETENCY ASSESSMENT
PROVE YOUR SKILLS.
EARN YOUR CONFIDENCE.

DEBRIEF & FEEDBACK
REFLECT. LEARN.
GET BETTER.

DEVELOPMENT PATH
PLAN YOUR GROWTH.
REACH YOUR GOALS.

CERTIFICATION READINESS
LEAVE PREPARED.
READY FOR THE FIELD.

COMPETENCY ASSESSMENT
Demonstrate independent skills.

DEBRIEF & LESSONS LEARNED
Open feedback.
Stronger performance.

PERSONAL DEVELOPMENT PLAN
Set goals.
Track progress.

CAREER ADVANCEMENT
Build experience.
Grow your future.

CERTIFICATION PREPAREDNESS
Meet standards.
Enter the field ready.

PROVE TODAY.
PREPARE FOR TOMORROW.
BE THE PROFESSIONAL.
SKILLS. JUDGMENT. GROWTH.

Assessment Day — Briefing, Written Test & Practical Evaluation

01

Assessment-day orientation

DISCUSS What happens today, in order?

02

Eligibility & documents

DISCUSS ID, affidavit, logbook — who has what?

03

Written test

DISCUSS Read every question twice

04

Practical evaluation

DISCUSS Plan, verify, move, re-verify

05

Grading & provisional results

DISCUSS How are P, D and F given?

06

Debrief & logbook closeout

DISCUSS What gets logged before you leave?

INSTRUCTOR NOTE The evaluator runs the day. Instructors brief, check documents and step back — no coaching once the evaluation starts.

Assessment-Day Orientation

WHAT / HOW

Assessment is independent evaluation

Verify eligibility and documents, explain assessment boundaries and let the assessor run the process per SPRAT requirements.

WHY

Why training and assessment are separated

Competence must be demonstrated without the instructor feeding the candidate each step.

WHEN / WHEN NOT

Use the SPRAT process

Every candidate takes a written test and a practical evaluation set for their SPRAT level.

WHAT GOES WRONG

Last-minute coaching during assessment

Prompting can compromise the independence and validity of the evaluation.

EVALUATOR'S EYE

Evaluator focus

Candidate arrives prepared, understands the rules and demonstrates their own competence.

TRAINEE Q&A

“Can my instructor help if I get stuck?”

During formal independent assessment, follow the evaluator’s rules; do not expect coaching as in training.

Theory & Verbal Knowledge

WHAT / HOW

Explain — not just recite

Candidates understand equipment, hazards, standards / SPRAT concepts, rigging, rescue and role responsibilities appropriate to their level.

WHY

Why verbal reasoning matters

Safe rope access requires decisions when the exact training setup is not present.

WHEN / WHEN NOT

Answer within your competence

Know when the correct answer is to stop and escalate.

WHAT GOES WRONG

Guessing technical limits

Inventing ratings, anchor capacities or requirements is worse than recognizing that authoritative information is needed.

EVALUATOR'S EYE

Evaluator focus

Accurate knowledge, sound reasoning and recognition of limits.

TRAINEE Q&A

“What if I don’t remember a number?”

Do not invent it. Explain where you would verify the controlling requirement.

Practical Assessment Behavior

WHAT / HOW

Plan → verify → move → re-verify

Use deliberate sequencing, connection checks, controlled movement and continuous redundancy.

WHY

Why safety overrides speed

Fast completion has no value if the system becomes uncontrolled or unsafe.

WHEN / WHEN NOT

Pause when needed

A deliberate safe pause to assess the next step beats rushing into a dangerous sequence.

WHAT GOES WRONG

Panic after a small error

Candidates compound a minor issue by moving faster or disconnecting the wrong component.

EVALUATOR'S EYE

Evaluator focus

Independent safe performance, correct technique, judgment and self-correction within SPRAT criteria.

TRAINEE Q&A

“Should I restart if I make a mistake?”

Follow the evaluator’s instructions. The priority is to stay safe and avoid compounding the error.

Results, Debrief & Remediation

WHAT / HOW

Turn the outcome into a development plan

Review the result, feedback, strengths, gaps and any SPRAT next steps.

WHY

Why the debrief matters

Passing is not mastery of every future scenario; an unsuccessful assessment should produce a focused improvement plan.

WHEN / WHEN NOT

Document accurately

Maintain training / assessment records and any required logbook entries.

WHAT GOES WRONG

Treating certification as the finish line

Skills decay, and work environments introduce hazards not seen in training.

EVALUATOR'S EYE

Evaluator focus

Professional response to feedback and a clear understanding of limitations and next steps.

TRAINEE Q&A

“If I pass, can I supervise immediately?”

Only within the authority of your certification level, employer / site requirements and demonstrated competence.

CLASSROOM → PRACTICAL

Day 5 Practical Training Plan

- Eligibility / equipment check and briefing
- Theory / verbal knowledge assessment
- Practical maneuvers by candidate level
- Rigging and rescue assessment by candidate level
- Results and debrief
- Logbook / recertification / development closeout



Photo: Christian Thiel · CC BY-SA 4.0 · Wikimedia Commons

What to Expect From Your Instructor

1 · Explain

Use the six teaching boxes on every lesson: what / how, why, when / when-not, failure modes, evaluator focus and trainee Q&A.

2 · Demonstrate

Show the complete technique once, then break it into decision points and safety checks.

3 · Coach

Correct unsafe sequence immediately; use questions so trainees explain the reason for the correction.

4 · Repeat

Progress from assisted practice to independent repetition without sacrificing redundancy or control.

5 · Evaluate

Observe the whole process — planning, setup, movement, self-checking, communication and recovery — not just the final position.

6 · Document

Record attendance, JSA participation, skill progression, remediation and readiness for assessment.

Document & Standards Verification Before Each Course

- ✓ Verify the current SPRAT syllabus and certification documents applicable to the candidates being trained.
- ✓ Verify applicable OSHA requirements for the actual work environment — do not present scheme guidance as a substitute for law.
- ✓ Verify manufacturer instructions for every harness, connector, rope, descender, ascender, backup device, pulley and rescue component.
- ✓ Confirm training structures, anchors and engineered systems are approved for the intended loads and exercises.
- ✓ Update lesson plans when official scheme documents, standards, manufacturer instructions or equipment change.
- ✓ Maintain a revision number, effective date, reviewer / approver and list of controlling documents for this instructor edition.

SOURCE NOTE

SPRAT content checked against Rope Access Certification Requirements CRC-01, v25A (January 2025). Re-verify at sprat.org before each course cycle.

SKYTECH ACADEMY

ROPE ACCESS TRAINING CENTER

OILFIELD | INDUSTRY | SAFETY | EXPERTISE

SAFETY THROUGH COMPETENCE.

Train the technician to understand the system — not merely memorize the maneuver.

REDUNDANCY • CONTINUOUS PROTECTION • ZERO COMPROMISE



KNOW THE SYSTEM
Understand how everything works together.

CONTROL THE RISK
Assess, plan and execute with confidence.

PROTECT EACH OTHER
Teamwork and communication save lives.

PERFORM EVERY TIME
Consistent skills. Consistent results.

BUILD YOUR FUTURE
Professional growth. Lifelong competency.

COMPETENT TODAY. PROFESSIONAL TOMORROW. SAFER TOGETHER.